



Design & Implementation

Only for course Teacher						
		Needs Improvement	Developing	Sufficient	Above Average	Total Mark
Allocate mark & Percentage		25%	50%	75%	100%	15
Understanding/Analysis	2					
Implementation	4					
Accuracy	2					
Task Efficiency	2					
Total obtained mark						
Comments						

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1. Introduction

This document presents the design and implementation details of the FarmFusion – Smart Agro Dairy Ecosystem, focusing on the UML and non-UML diagrams developed to represent the system structure, behavior, data flow, architecture, and user interaction. The document provides a detailed explanation of the major diagrams used during the system analysis and design phase.

The diagrams illustrate how Administrators, Staff, Veterinarians, Farmers, Wholesalers/Customers, and automated environmental services interact with the FarmFusion platform. The major system functionalities include user registration and login, animal health monitoring, automated inventory management, feeding and nutrition management, dairy product management, dairy product wholesale management, crop harvesting, sales/billing/expense tracking, and weather monitoring with alerts.

The design documentation is prepared to provide a clear understanding of the system before and during implementation. FarmFusion is intended to make farm operations more connected, organized, traceable, and efficient by providing a centralized digital ecosystem for daily agricultural decisions and operational management.

2. UML

This chapter presents the UML diagrams developed for the FarmFusion – Smart Agro Dairy Ecosystem. These diagrams describe the structural and behavioral aspects of the system and show how different users and system components interact with the major farm-management functions. The UML design includes the Use Case Diagram, Class Diagram, Activity Diagrams, Sequence Diagrams, and State Diagrams. Together, these diagrams provide a complete representation of the system's functionality, structure, interactions, workflows, and state transitions.

Using UML during the design stage helps communicate the system requirements clearly and provides a strong foundation for implementation.

2.1 Use Case Diagram

The Use Case Diagram represents the major functional requirements of FarmFusion and shows how different actors interact with the system. The main actors identified in the system are Administrator, Staff, Veterinarian, Farmer, Wholesaler/Customer, and automated weather/environmental services.

The Administrator manages users, operational records, inventory, dairy products, crop operations, financial records, alerts, and system monitoring. Staff perform day-to-day operational activities such as inventory, dairy and crop-related tasks according to their permissions. Veterinarians focus on animal health records and monitoring. Farmers use the platform to monitor farm operations, crops, dairy, inventory, financial information and weather alerts. Wholesalers/Customers participate in product ordering and wholesale workflows, while automated environmental services provide weather data and alert triggers.

The Use Case Diagram also represents include and extend relationships where supporting processes are required by a major operation, such as authentication before accessing protected modules and alert generation when monitored conditions cross defined thresholds.

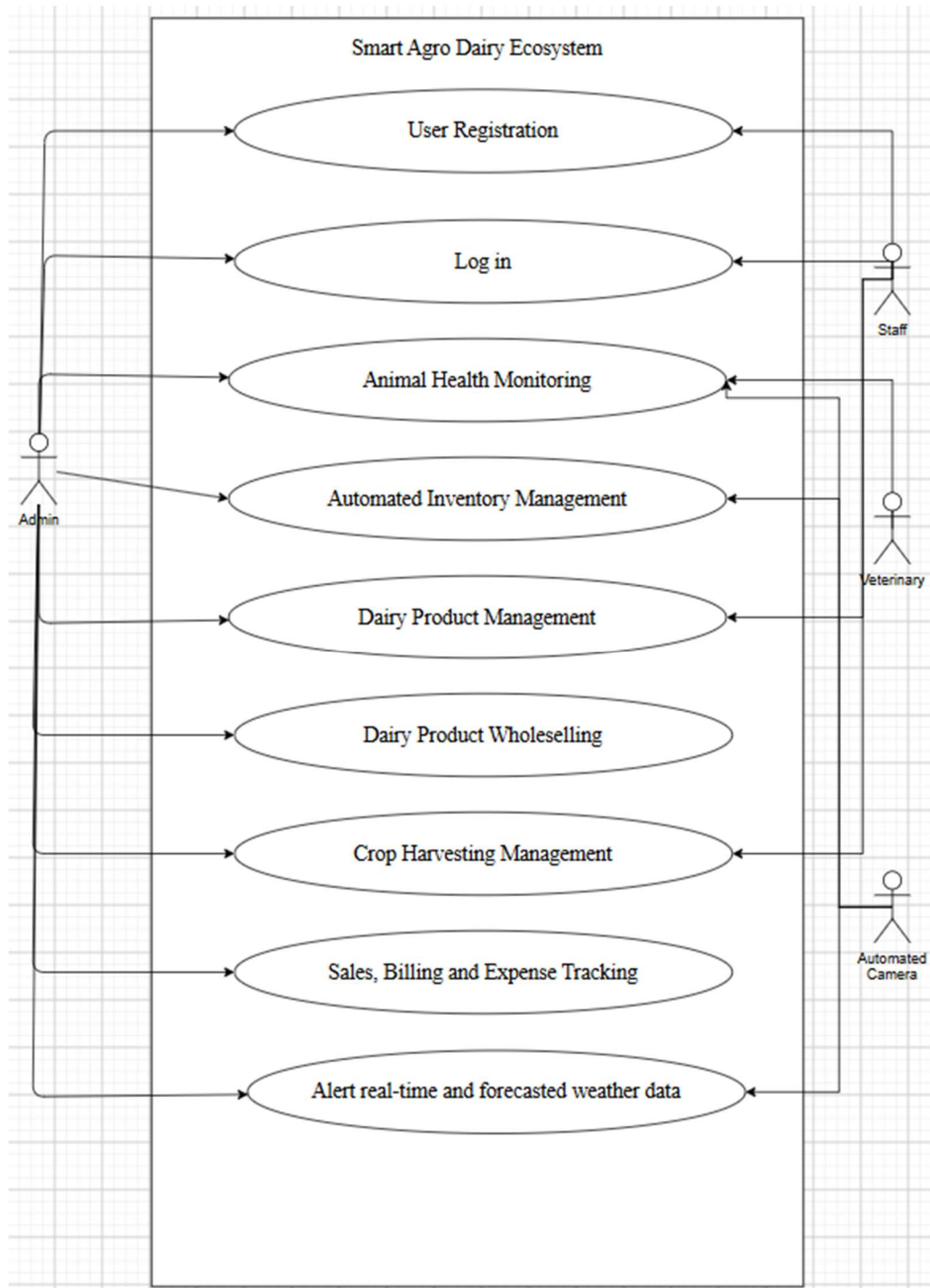


Figure 2.1.1: Use Case Diagram of the FarmFusion – Smart Agro Dairy Ecosystem

The Use Case Diagram (Figure 2.1.1) shows the main actors and their interactions with FarmFusion, including authentication, animal health, inventory, dairy operations, wholesale workflows, crop harvesting, sales and finance, and weather alert services.

2.2 Class Diagram

The Class Diagram represents the major classes of the FarmFusion system and the relationships between them. It provides a structural view of the system by showing important entities, attributes, operations, and relationships.

The core classes include User, Farmer, Veterinarian, Staff, Admin, Livestock, HealthRecord, Medicine/Vaccine, InventoryItem, FeedingPlan, FeedingSession, Crop, HarvestRecord, DairyProduct, WholesaleOrder, SalesOrder, Invoice, Payment, Expense, WeatherData, Alert, and related operational records. These classes are connected through relationships that represent how users manage farm resources and how operational events create or update system records.

The User and role-specific classes provide controlled access to the platform. Livestock and HealthRecord support veterinary workflows, while InventoryItem and FeedingSession support stock and feeding operations. Crop and HarvestRecord manage cultivation and harvesting. DairyProduct and WholesaleOrder support dairy commerce. SalesOrder, Invoice, Payment, and Expense provide the financial workflow. WeatherData and Alert connect environmental monitoring with operational notifications.

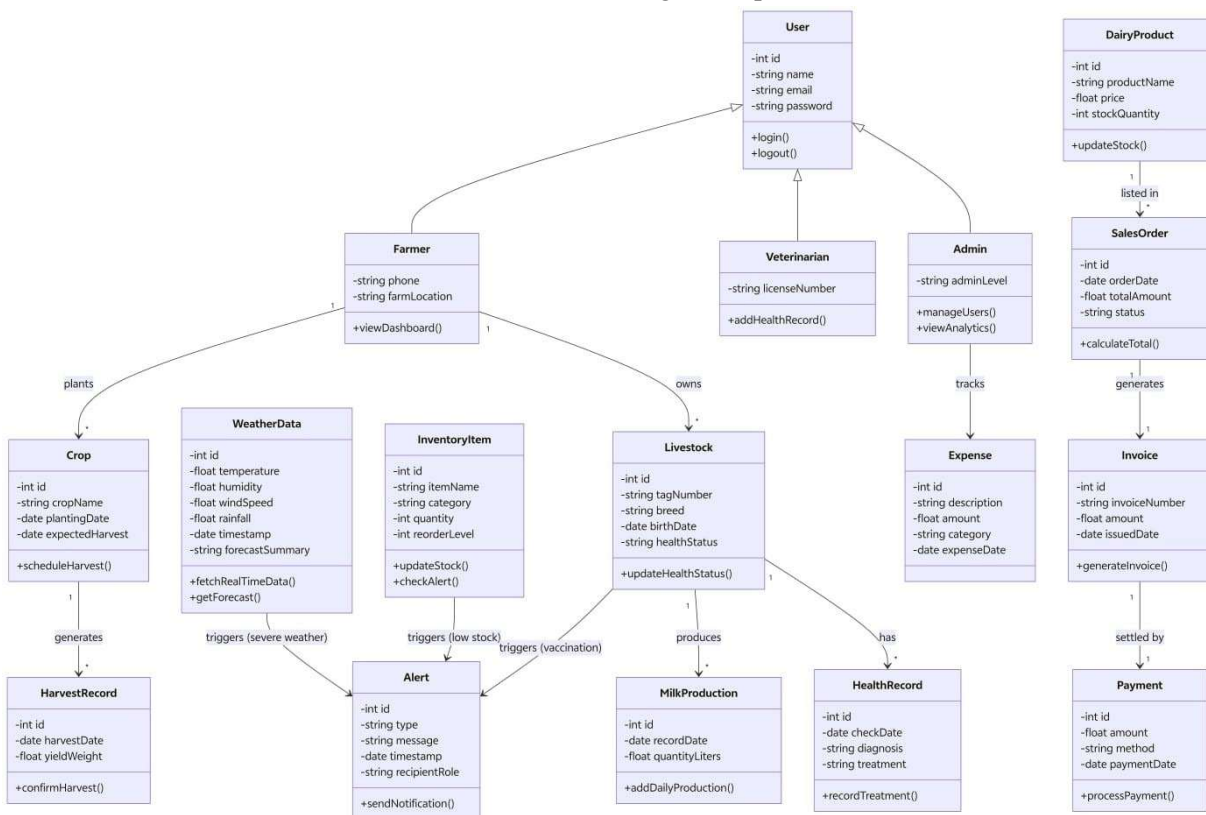


Figure 2.2.1: Class Diagram of the FarmFusion – Smart Agro Dairy Ecosystem

The Class Diagram (Figure 2.2.1) shows the principal classes, attributes, methods, and relationships of FarmFusion. It represents the overall structure of the system and explains how different classes interact to perform farm-management operations.

2.3 Activity Diagrams

The Activity Diagrams represent the workflow of the major functionalities of FarmFusion. They provide a visual representation of the sequence of activities performed by users and the system, including inputs, system responses, decision points, validation, alternative flows, and final outcomes.

The Activity Diagrams cover the major functions of the system, including user registration, login, animal health monitoring, automated inventory and feeding, dairy product management, dairy product wholesale management, crop harvesting, sales/billing/expense tracking, and weather monitoring and alert generation. These diagrams help to understand how each functionality works from the starting point to the final outcome.

1.User Registration:

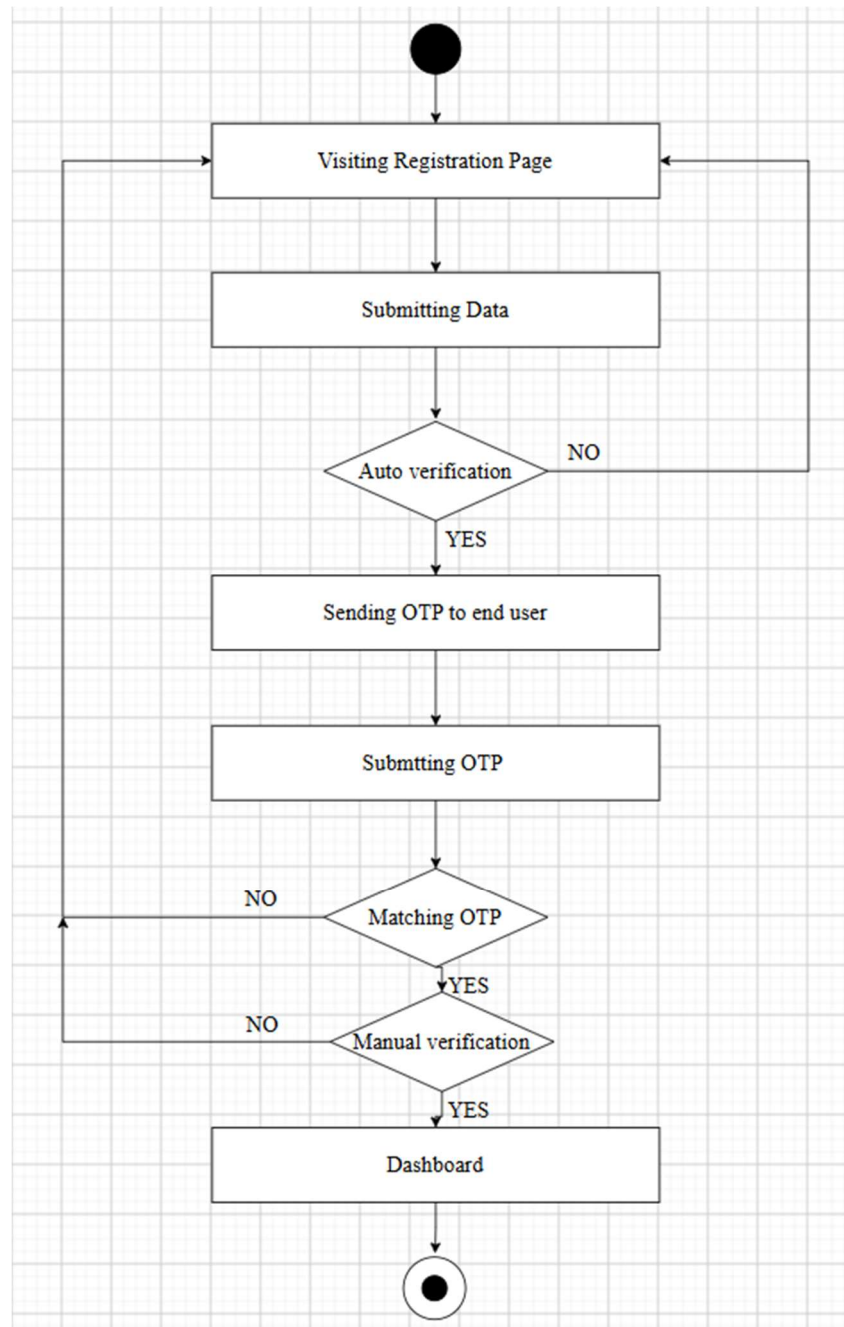


Figure 1: Activity Diagram for User Registration

The activity starts when a user opens the registration page and provides the required profile information. The system validates the information, checks for duplicate account information, creates the account when valid, and confirms successful registration.

2.User Login:

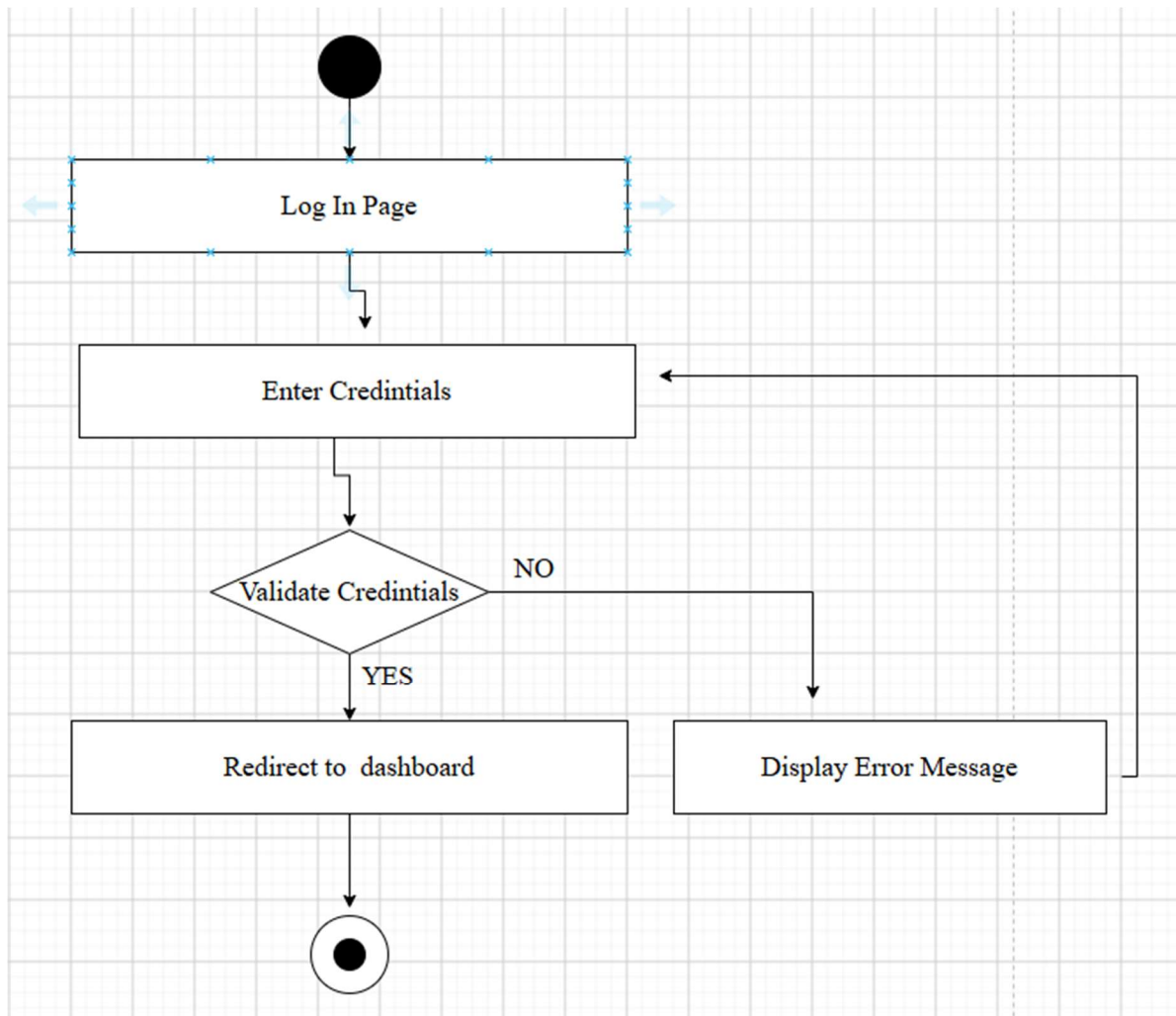


Figure 2: Activity Diagram for User Login

The workflow begins at the login page. The user provides credentials, the system validates them, and valid credentials lead to the appropriate role dashboard while invalid credentials produce an error and allow another attempt.

3. Animal Health Monitoring:

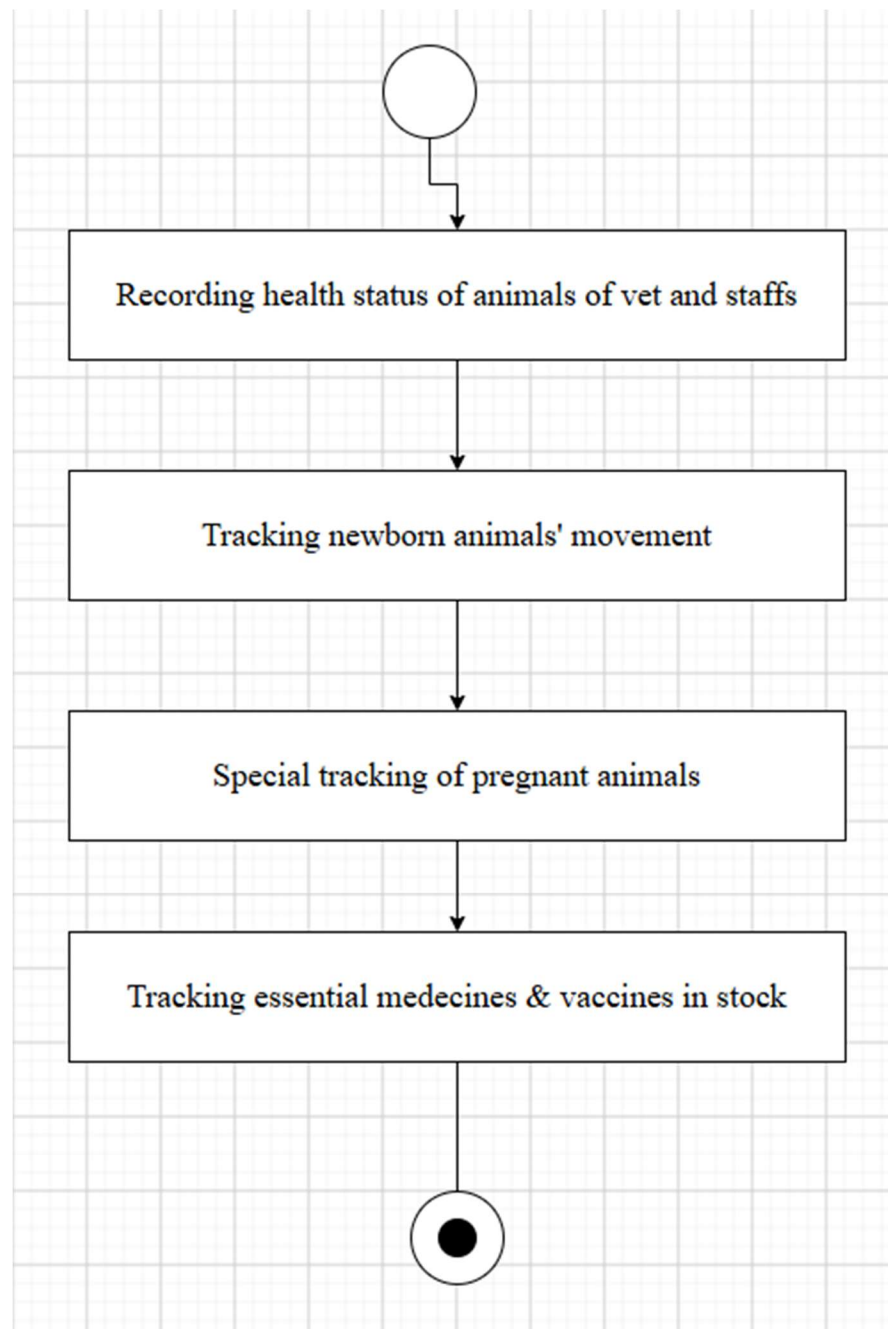


Figure 3: Activity Diagram for Animal Health Monitoring

The veterinarian or authorized user opens the animal-health workspace, selects or records animal information, updates health conditions, records treatment or monitoring information, and saves the updated health record. The system maintains the animal history for later review.

4. Automated Inventory and Feeding:

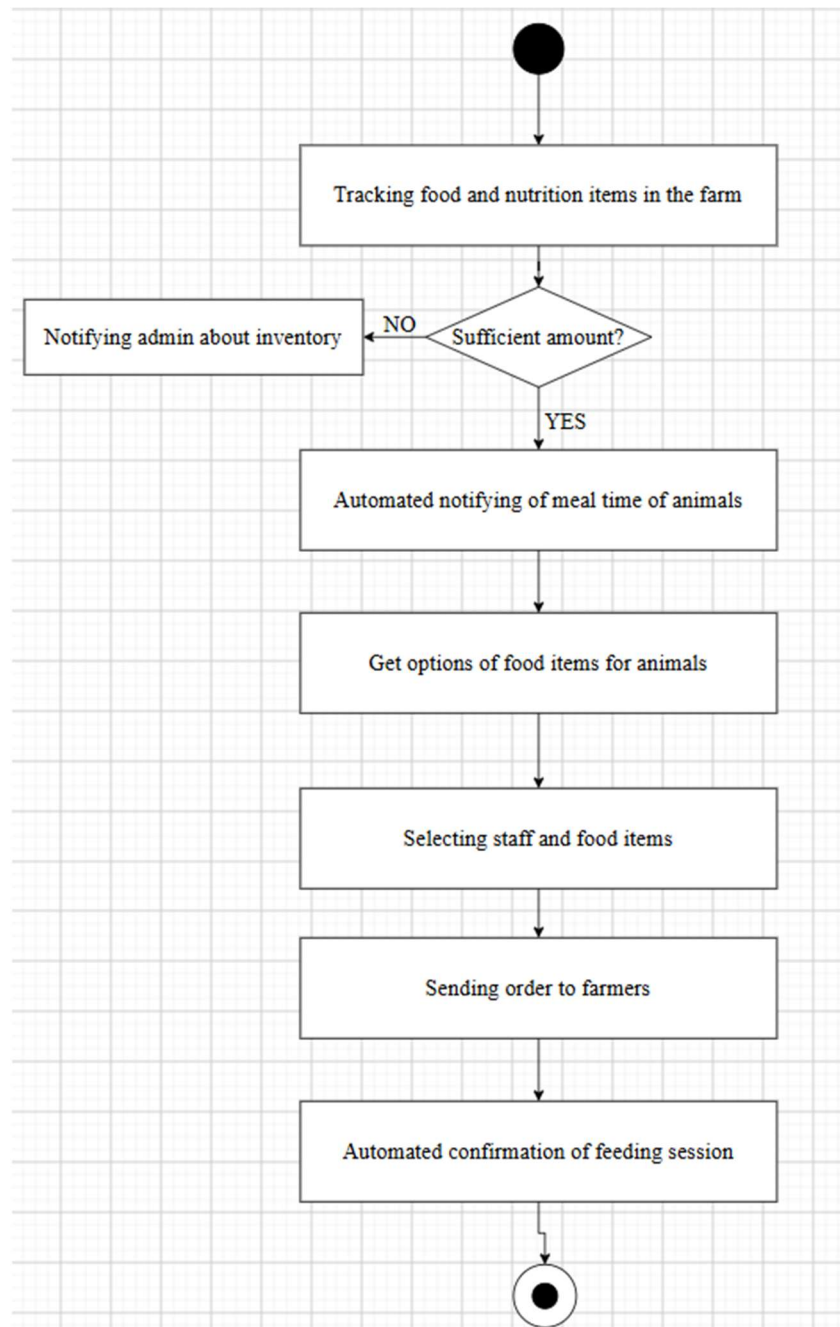


Figure 4: Activity Diagram for Automated Inventory and Feeding

The system checks stock levels and feeding requirements. If stock is below the required threshold, a low-stock alert is generated. Otherwise, the feeding session is scheduled, food and responsible staff are selected, the feeding order is issued, and the session is confirmed.

5.Dairy Product Management:

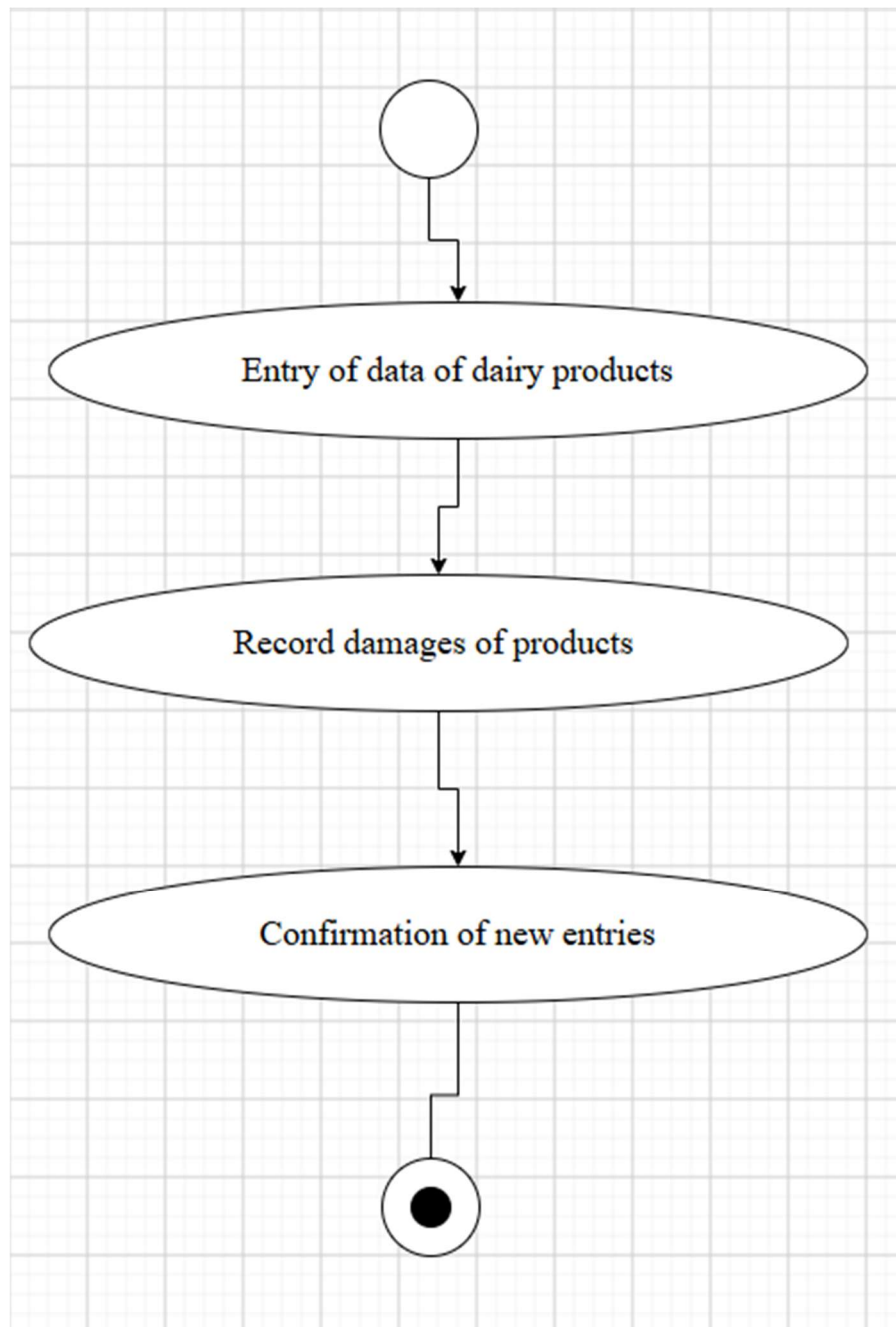


Figure 5: Activity Diagram for Dairy Product Management

The user opens dairy operations, records product information and quantity, checks damage or quality information, and confirms the entry. The system updates the dairy-product stock and makes the new record available to authorized users.

6.Dairy Product Wholesale Management:

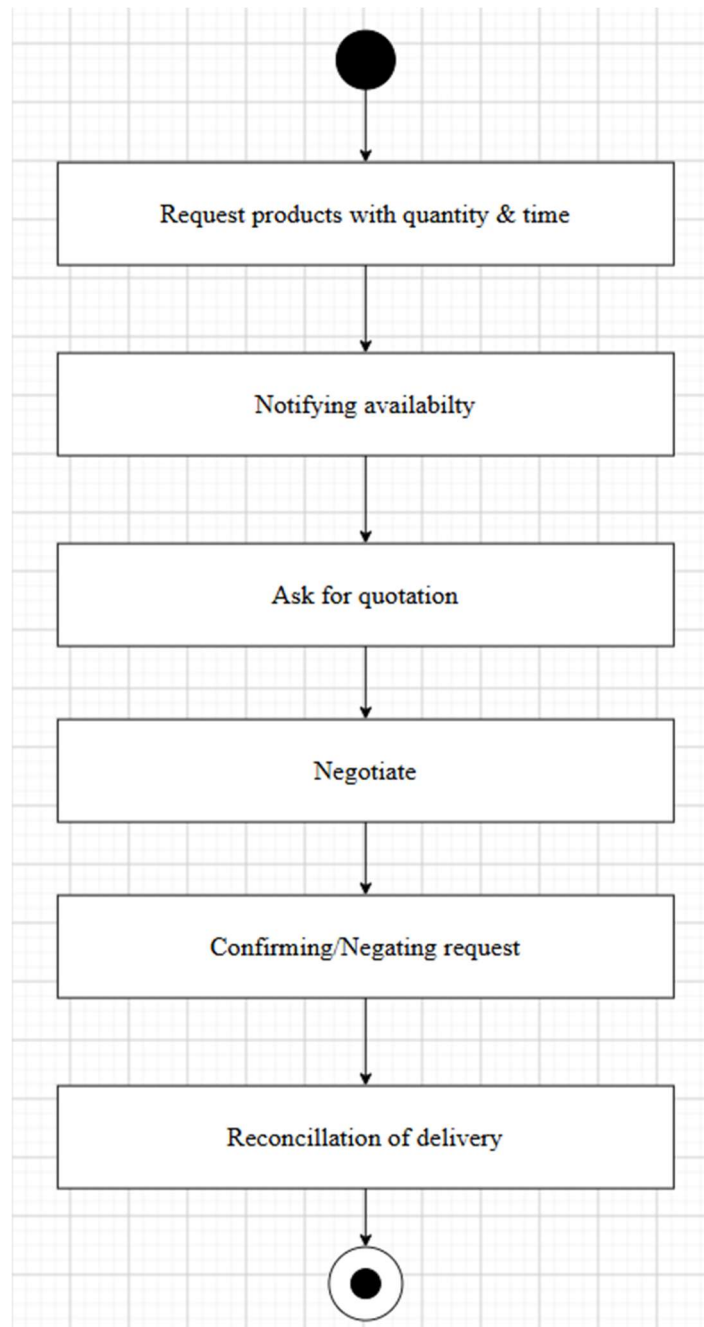


Figure 6: Activity Diagram for Dairy Product Wholesale Management

A wholesaler/customer requests a product. The system checks availability, prepares or receives a quotation, supports negotiation, confirms or rejects the request, and reconciles the delivered quantity with the confirmed order.

7.Crop Harvesting Management:

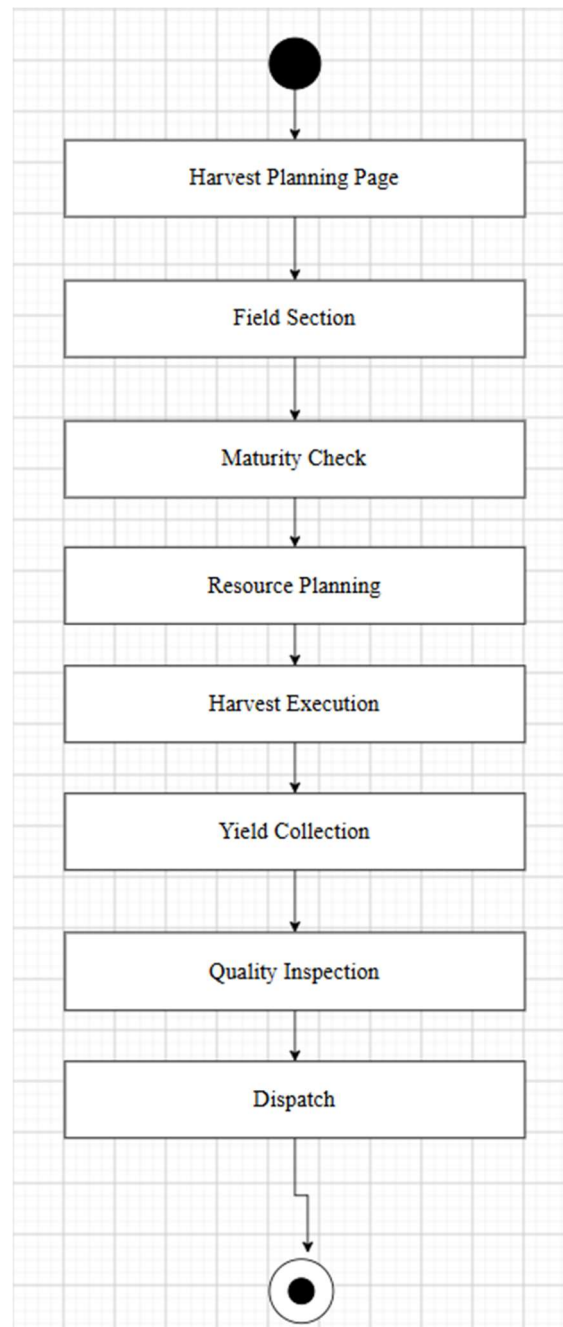


Figure 7: Activity Diagram for Crop Harvesting Management

The user selects a crop/field, checks maturity, plans resources, performs harvesting, records yield and quality information, and dispatches the harvested crop when the workflow is complete.

8.Sales, Billing and Expense Tracking:

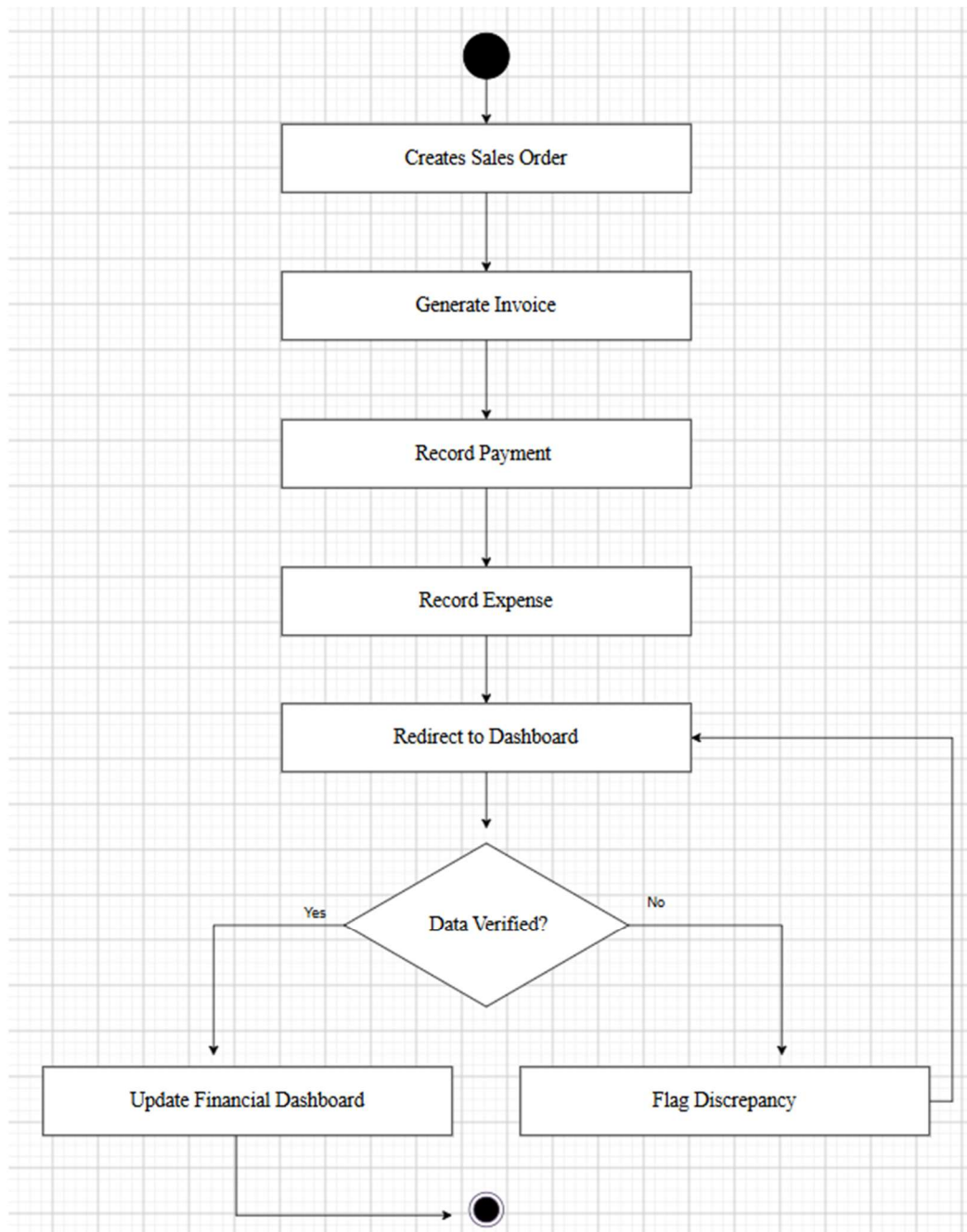


Figure 8: Activity Diagram for Sales, Billing and Expense Tracking

A sales order is created, the system calculates the amount and generates an invoice, payment is recorded, expenses are recorded, and the financial information is verified before the dashboard and reports are updated.

9. Weather Monitoring and Alert:

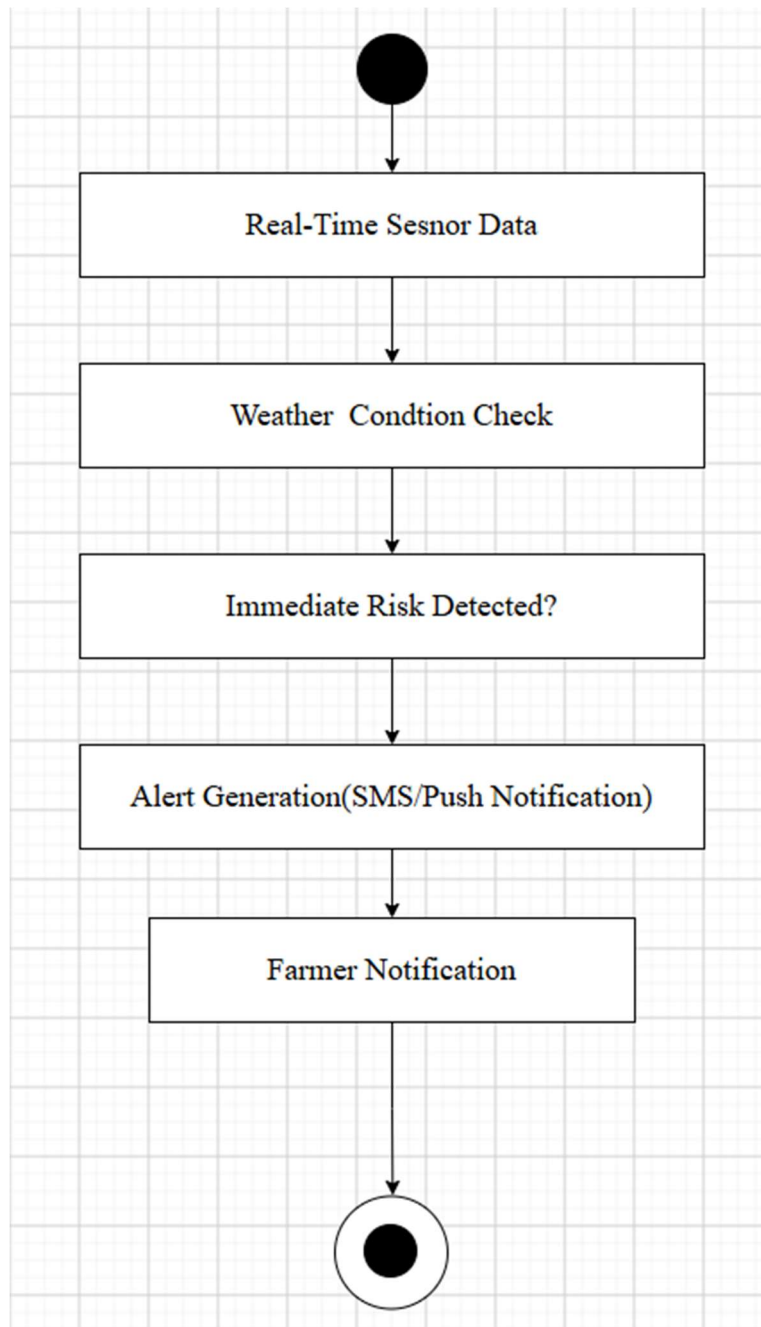


Figure 9: Activity Diagram for Weather Monitoring and Alert

The system receives environmental data, evaluates conditions, detects operational risk when necessary, generates an alert, and notifies the responsible users.

2.4 Sequence Diagrams

The Sequence Diagrams illustrate the step-by-step interaction between users, the FarmFusion application, databases, role services, operational modules, and automated environmental services. These diagrams show the order in which requests, validations, database operations, responses, and notifications occur during different system processes.

1. User Registration:

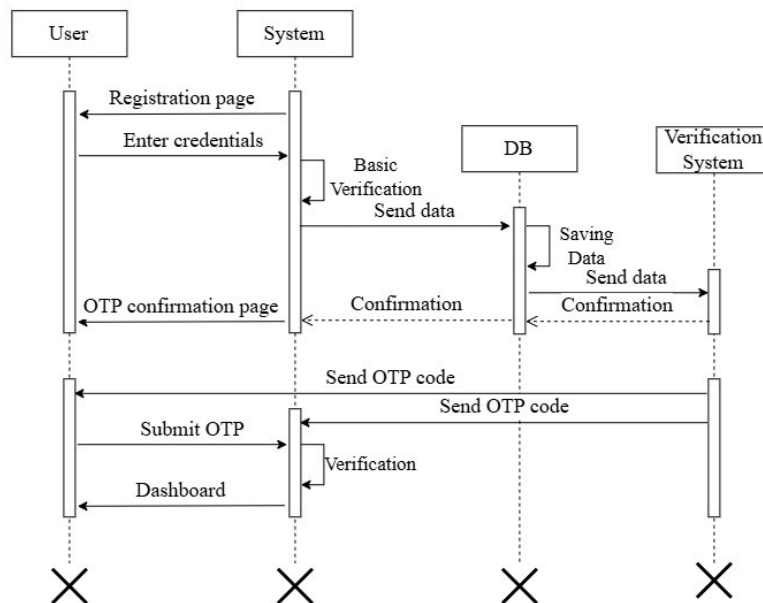
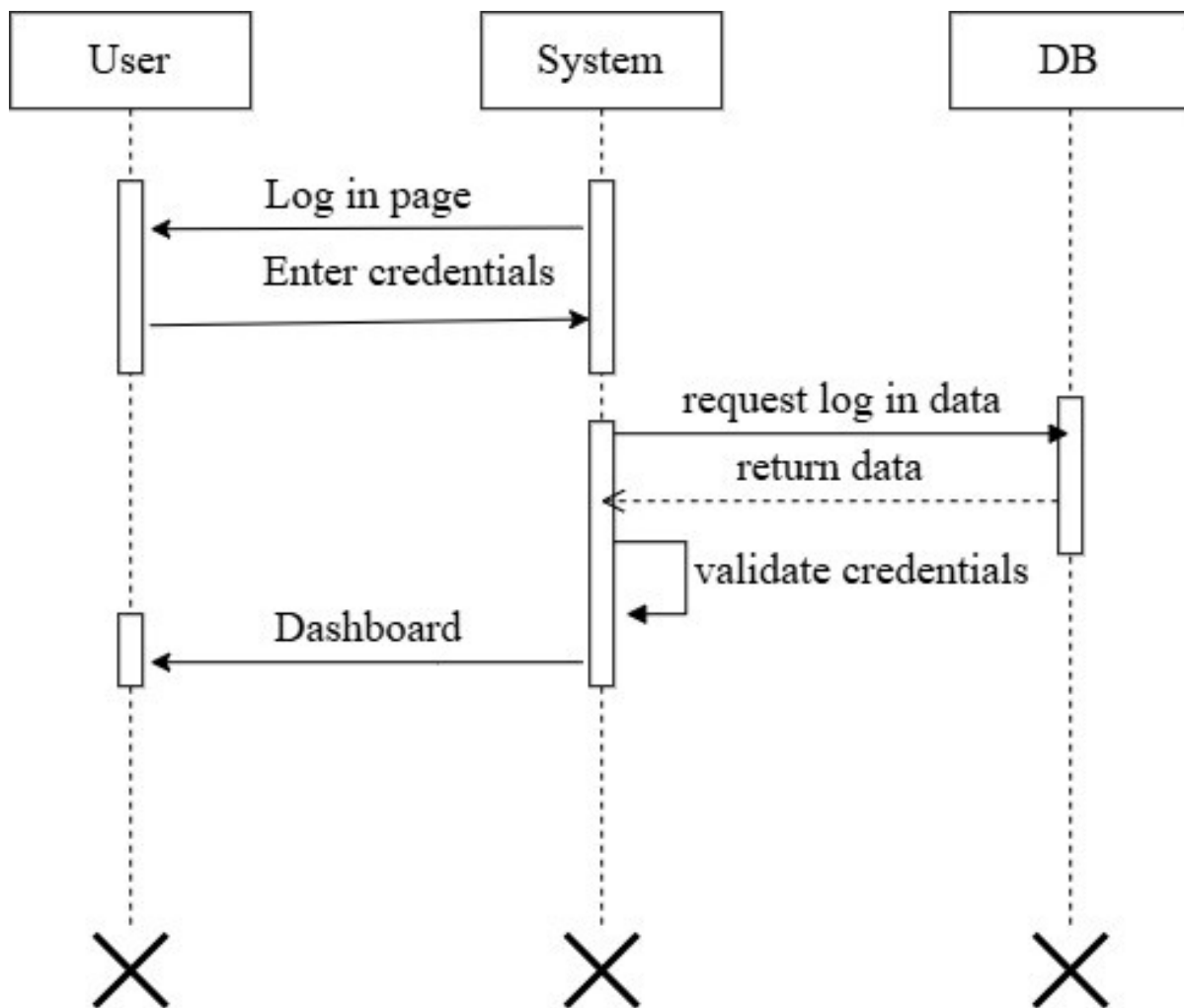


Figure 1: Sequence Diagram for User Registration

The sequence shows the user submitting registration information to the frontend, the authentication service validating the data, the database checking uniqueness and storing the new account, and the frontend displaying confirmation.

2.User Login:**Figure 2: Sequence Diagram for User Login**

The sequence shows credential submission, authentication validation, role identification, session creation, and dashboard redirection. Invalid credentials return an error response.

3. Animal Health Monitoring:

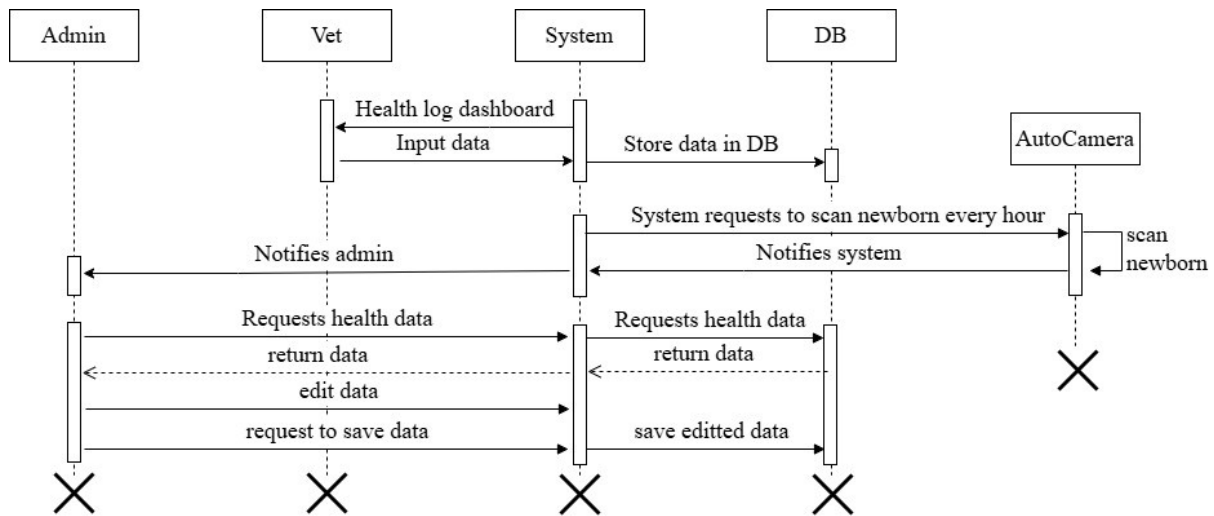


Figure 3: Sequence Diagram for Animal Health Monitoring

The veterinarian or authorized user selects an animal, submits health information, the system validates and stores the health record, and the updated health status is returned to the dashboard.

4. Automated Inventory Management:

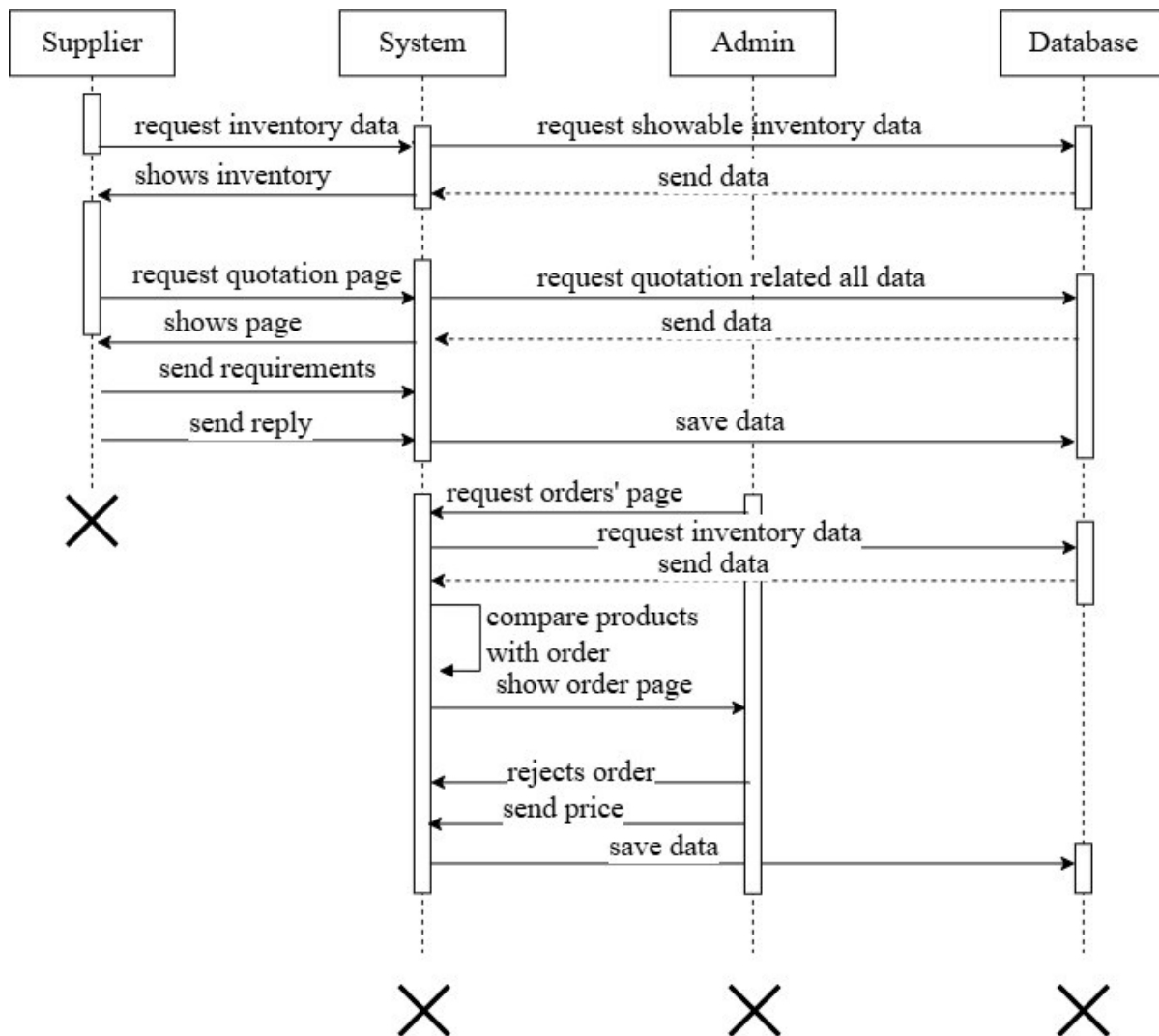


Figure 4: Sequence Diagram for Automated Inventory Management

The inventory module checks the current quantity against the reorder threshold, updates stock records, and generates a low-stock notification when the threshold is crossed.

5. Feeding and Nutrition Management:

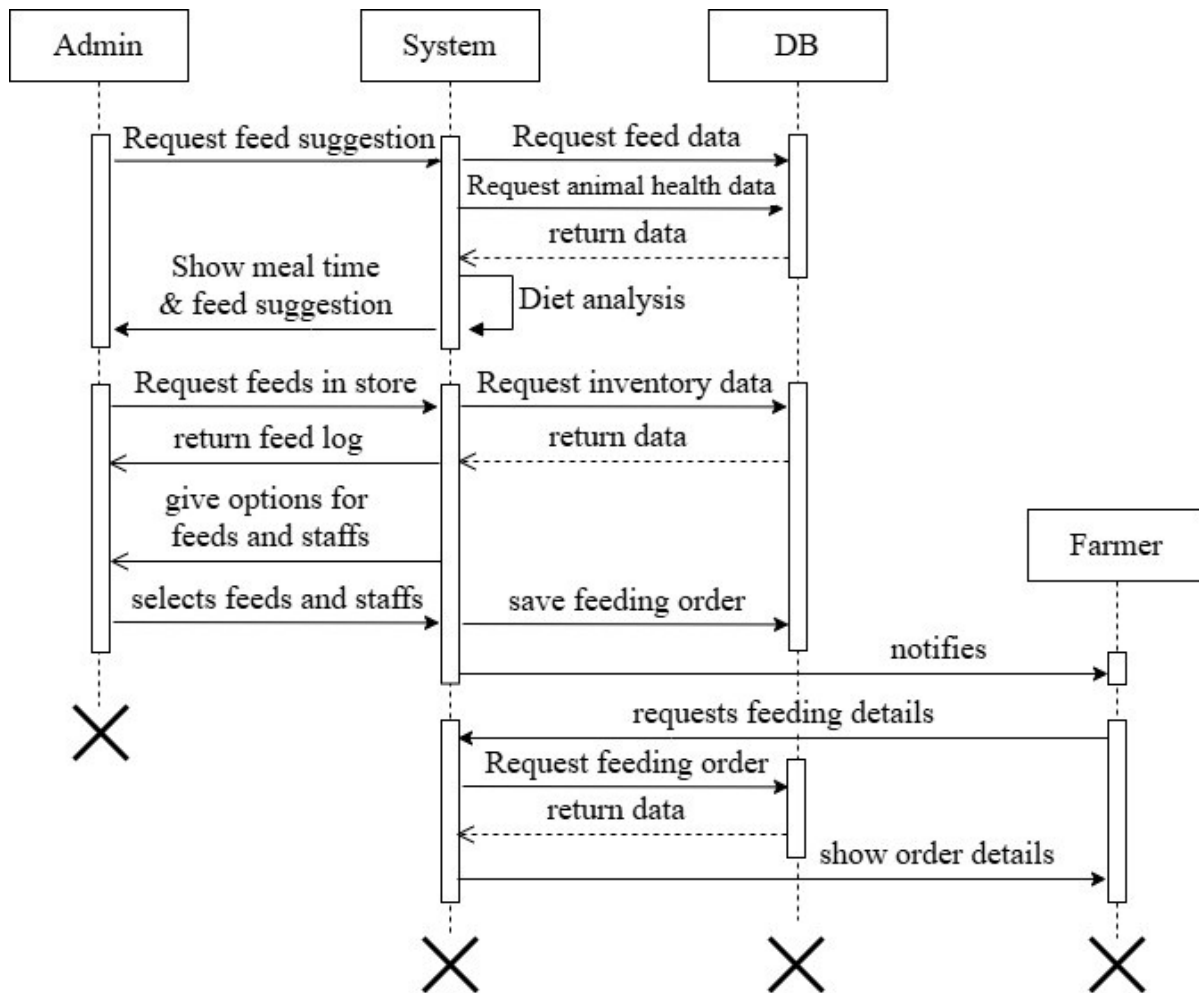


Figure 5: Sequence Diagram for Feeding and Nutrition Management

The system checks feed availability, selects the feeding plan and responsible staff, creates the feeding order, and records confirmation after the feeding session.

6.Crop Harvesting Management:

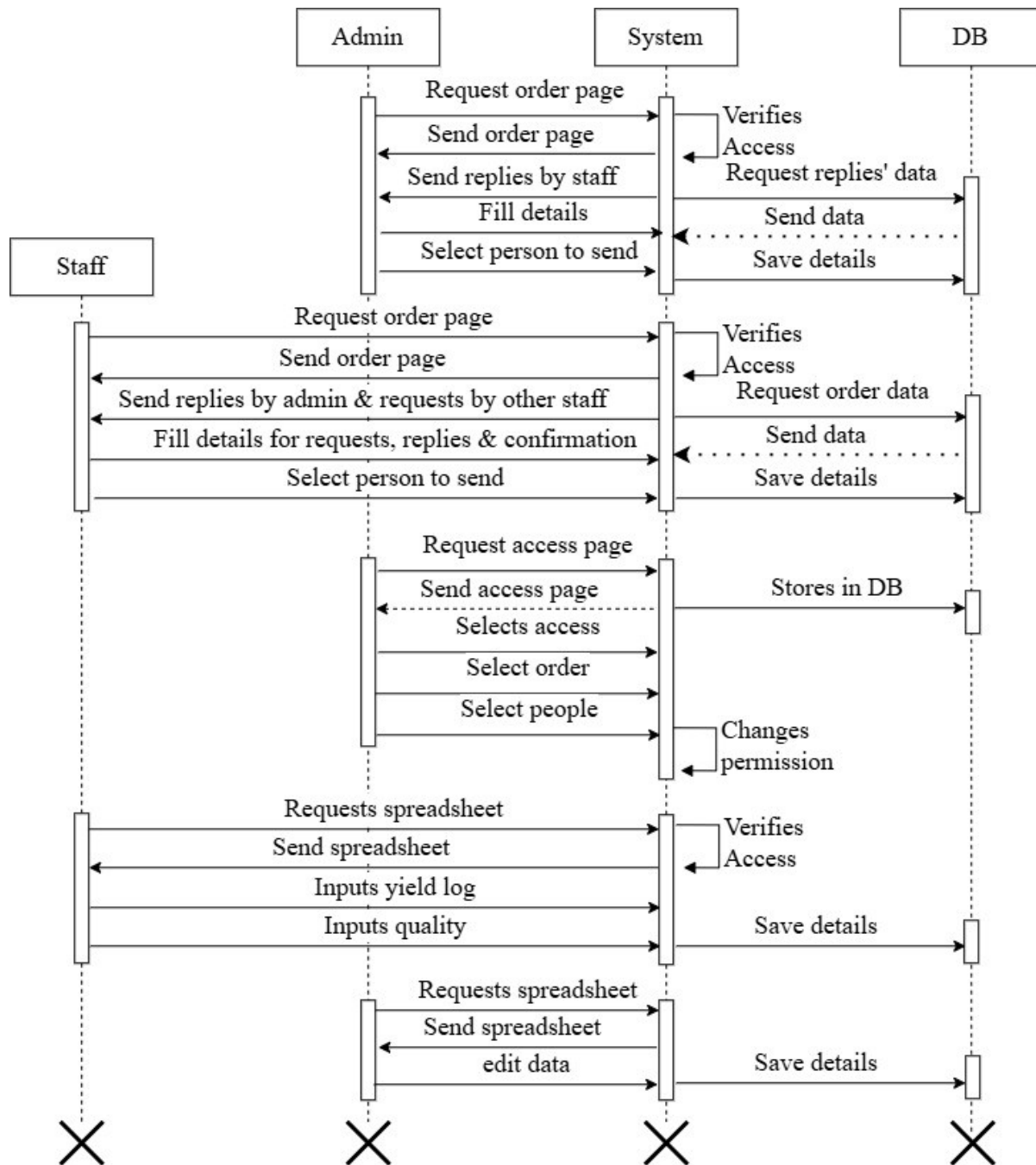


Figure 6: Sequence Diagram for Crop Harvesting Management

The system receives harvest planning data, checks maturity and resources, records the harvest, stores yield and quality information, and updates dispatch status.

7.Dairy Product Management:

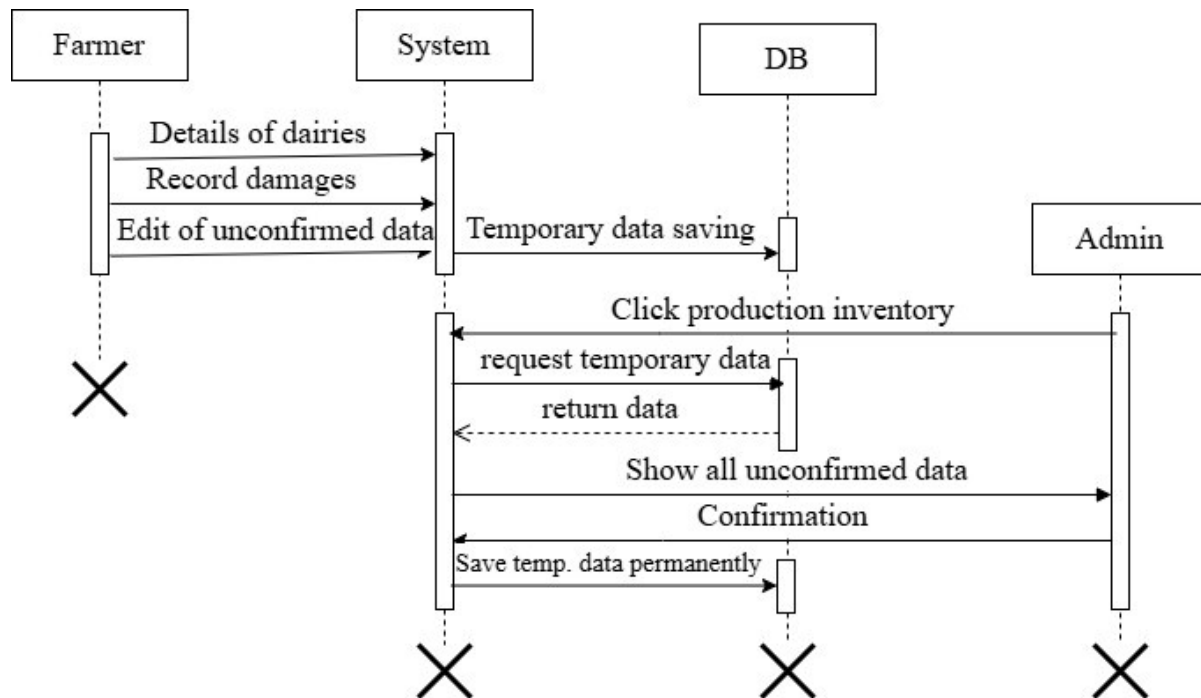


Figure 7: Sequence Diagram for Dairy Product Management

The user submits dairy-product information, the system validates and stores the product record, updates quantity and stock, and returns confirmation.

8.Weather Monitoring and Alert:

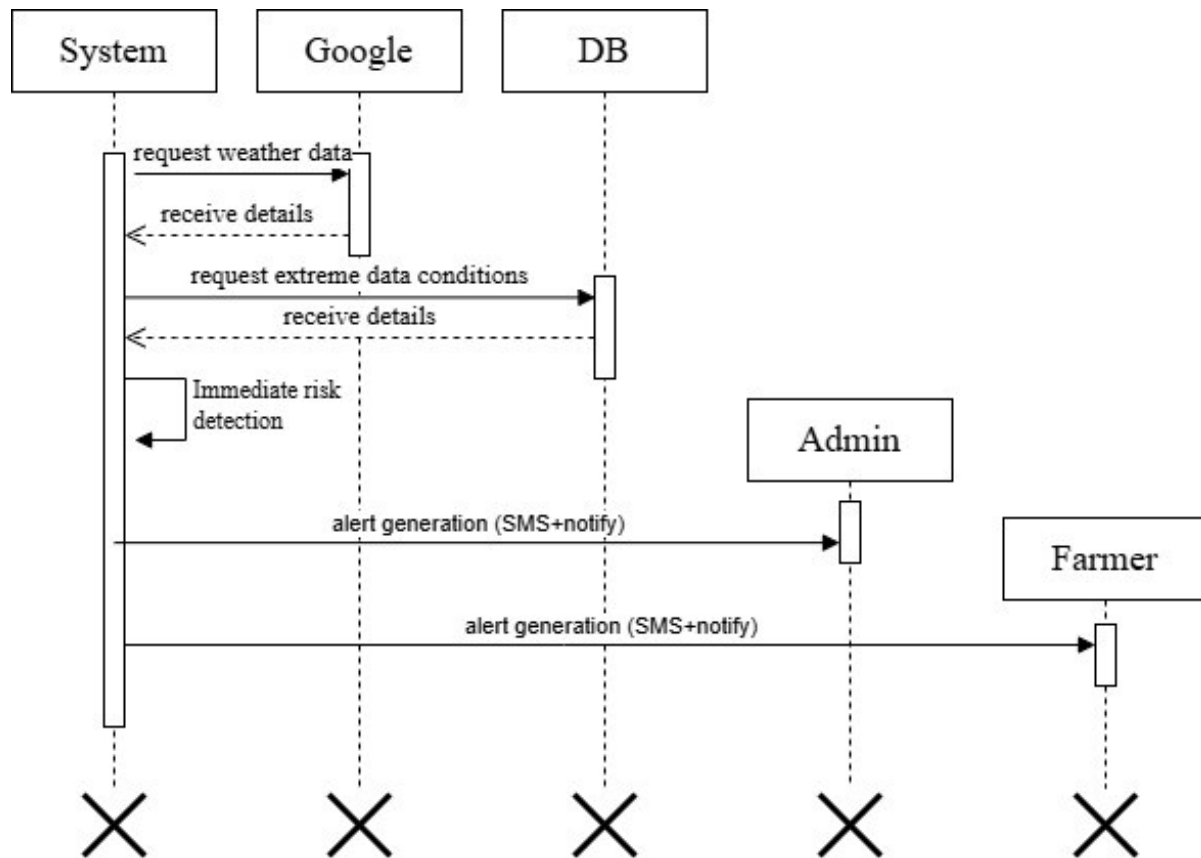


Figure 8: Sequence Diagram for Weather Monitoring and Alert

The environmental service sends weather data, the system processes the conditions, stores the data, evaluates risk, creates an alert if required, and notifies responsible users.

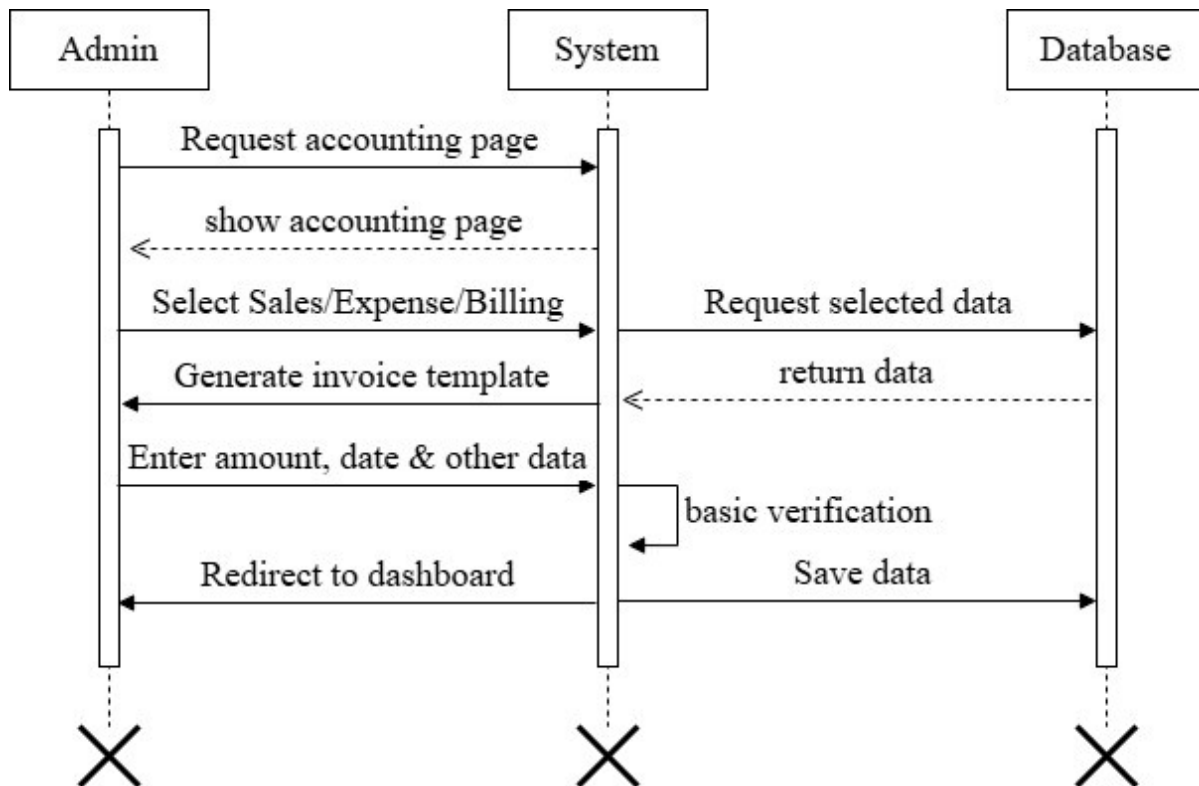
9.Sales, Billing and Expense Tracking:

Figure 9: Sequence Diagram for Sales, Billing and Expense Tracking

The user creates a sale, the system generates an invoice, records payment and expenses, verifies the transaction and updates financial summaries.

2.5 State Diagrams

The State Diagrams represent the different states through which important entities or processes of the FarmFusion system pass during their lifecycle. These diagrams show how system events and decisions cause an entity or workflow to move from one state to another.

1.User Registration:

FarmFusion - User Registration State Diagram

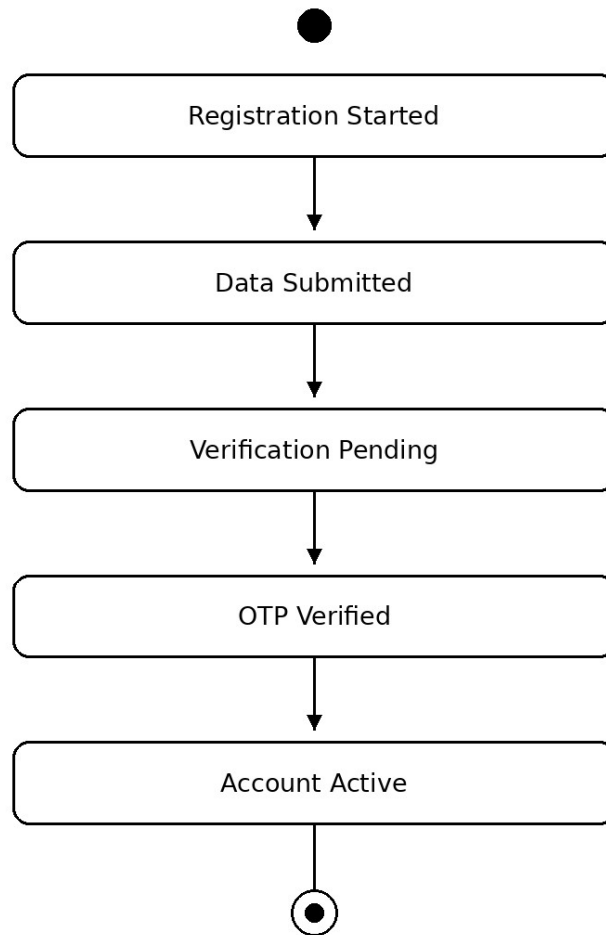


Figure 1: State Diagram for User Registration

The registration lifecycle moves from idle to registration form, validation, verification, account creation, and account activation. Invalid information or duplicate account details return the process to an error state before another attempt.

2.User Login:

FarmFusion - User Login State Diagram

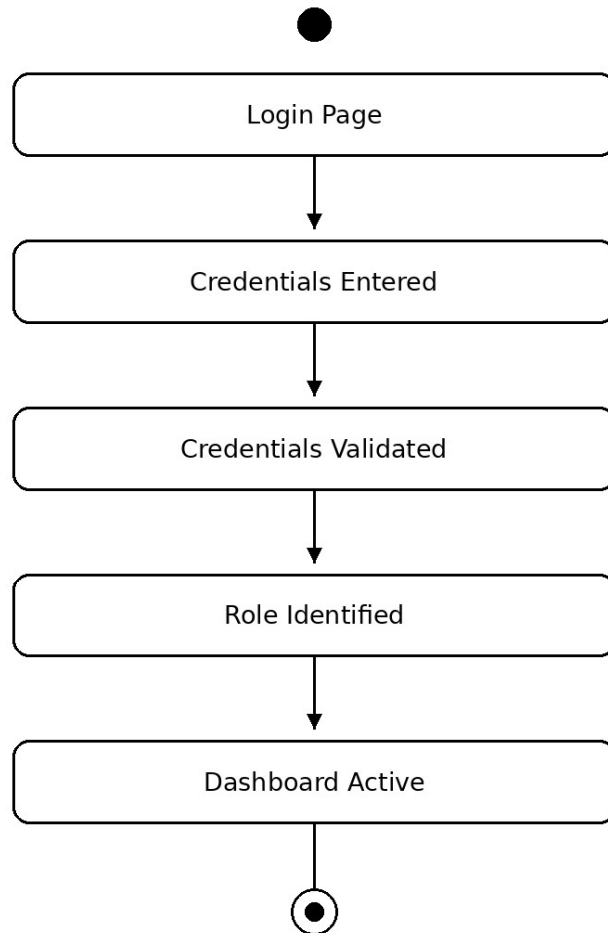


Figure 2: State Diagram for User Login

The login lifecycle moves from login page to credential entry and validation. Invalid credentials return the process to retry, while valid credentials move to authenticated and dashboard-active states.

3. Animal Health Monitoring:

nFusion - Animal Health Monitoring State Diag

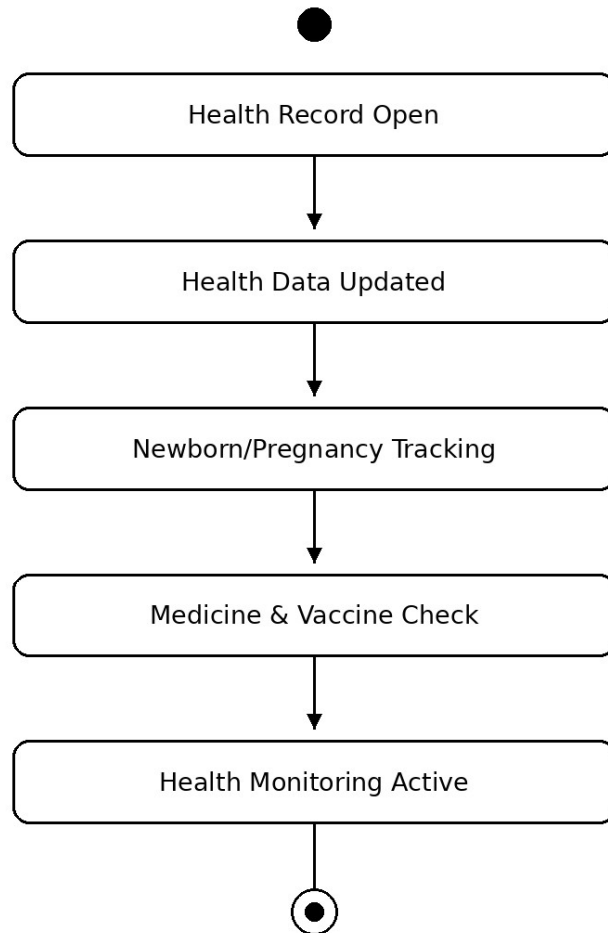


Figure 3: State Diagram for Animal Health Monitoring

The health workflow moves from an open record to data entry and validation, then to updated health status. Follow-up, treatment, newborn, pregnancy, medicine, or vaccine actions can create further monitoring states.

4. Automated Inventory Management:

ion - Automated Inventory Management State

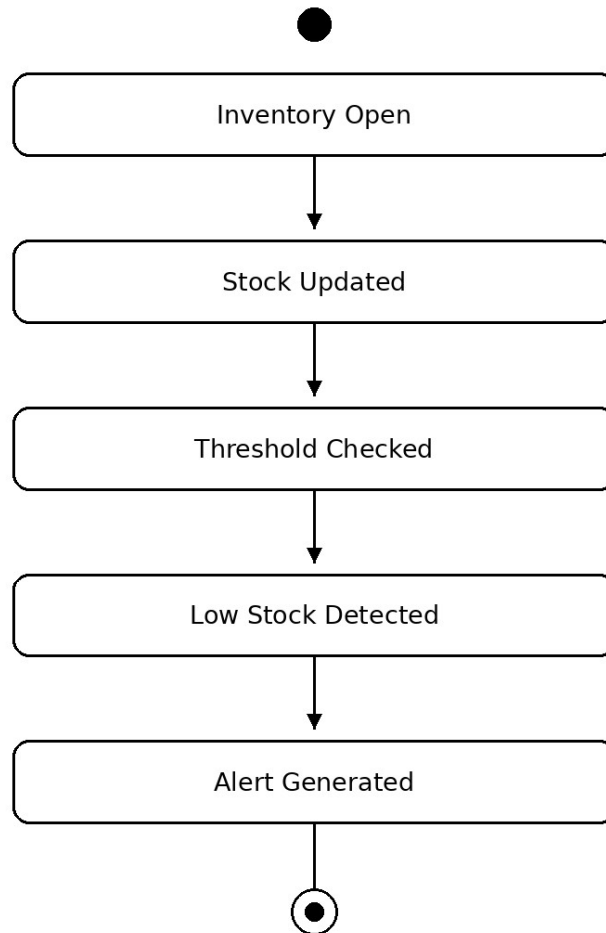


Figure 4: State Diagram for Automated Inventory Management

Inventory moves from open/update to threshold checking. When the stock falls below the reorder level, the system enters a low-stock state and generates an alert before returning to monitored stock.

5.Dairy Product Management:

1 Fusion - Dairy Product Management State Dia

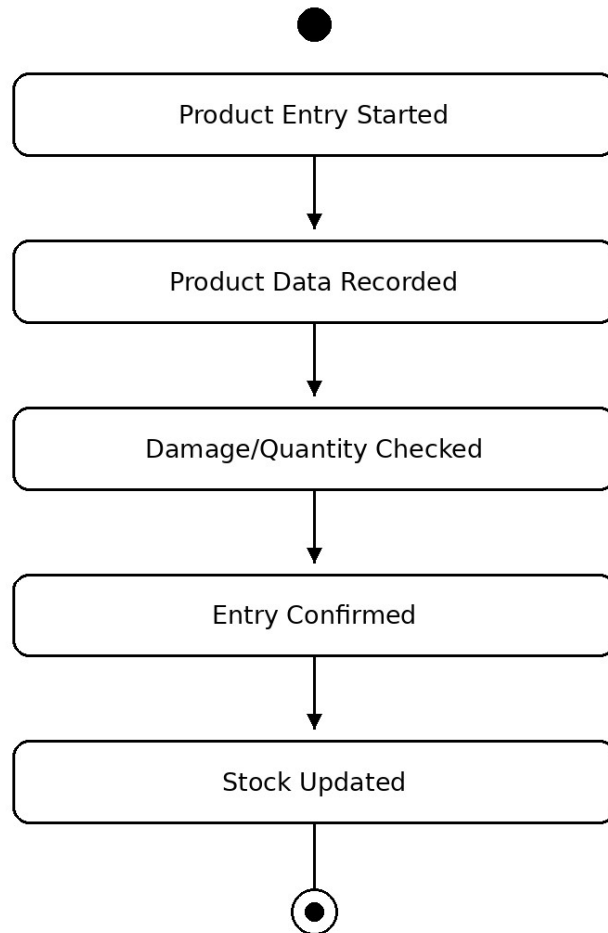


Figure 5: State Diagram for Dairy Product Management

A dairy product entry moves from creation to data recording, quantity/damage checking, confirmation, and stock-updated states.

6.Dairy Product Wholesale Management:

n - Dairy Product Wholesale Management Stat

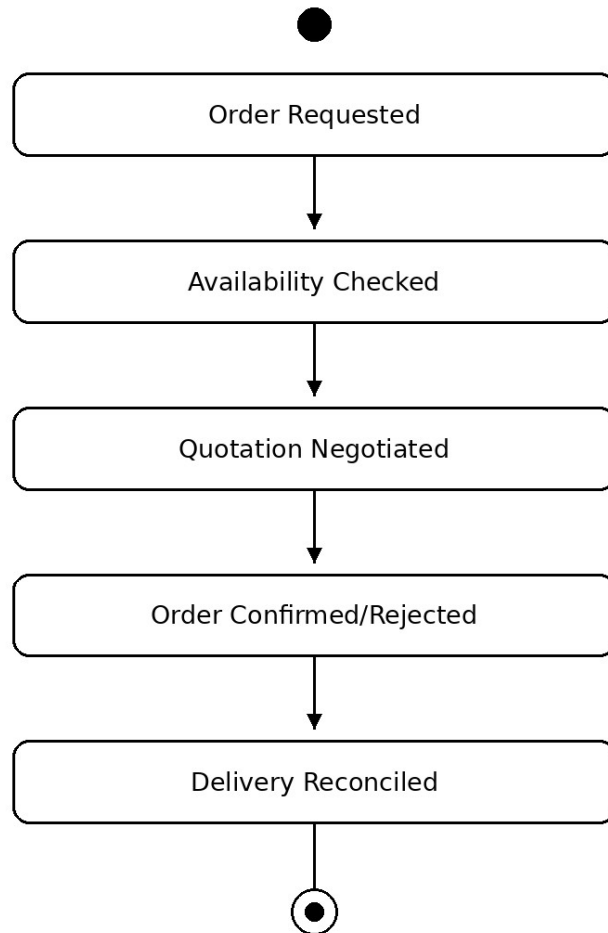


Figure 6: State Diagram for Dairy Product Wholesale Management

A wholesale request moves from requested to availability checked, quotation, negotiation, confirmed/rejected, and delivery reconciliation states.

7.Crop Harvesting Management:

Figure 7: State Diagram for Crop Harvesting Management

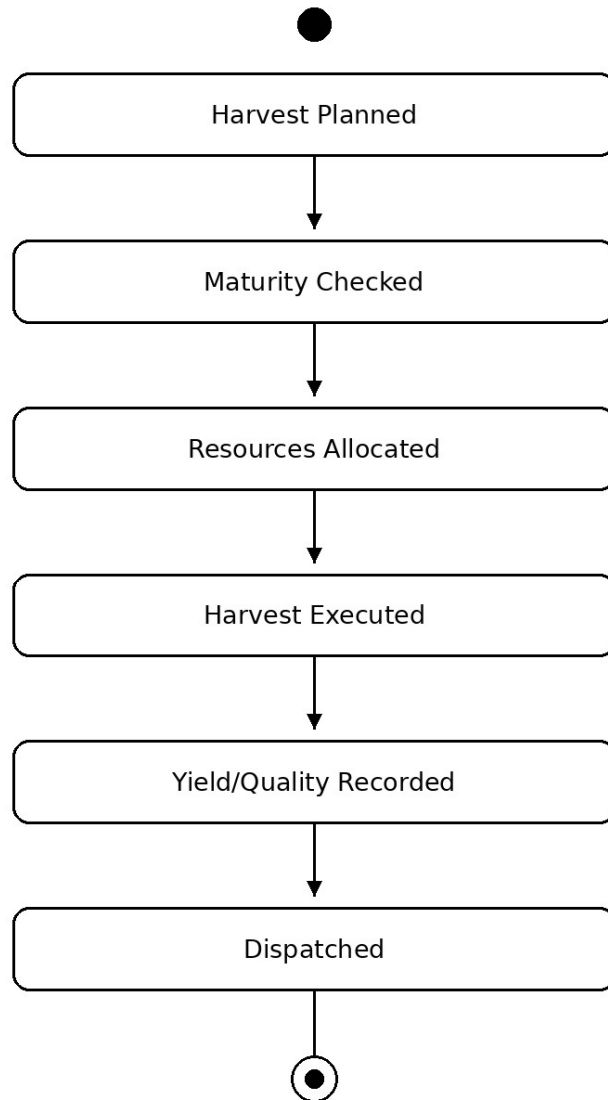


Figure 7: State Diagram for Crop Harvesting Management

The crop lifecycle moves from harvest planning to maturity checking, resource allocation, harvesting, yield/quality recording, and dispatch.

8.Sales, Billing and Expense Tracking:

sion - Sales, Billing & Expense Tracking State [

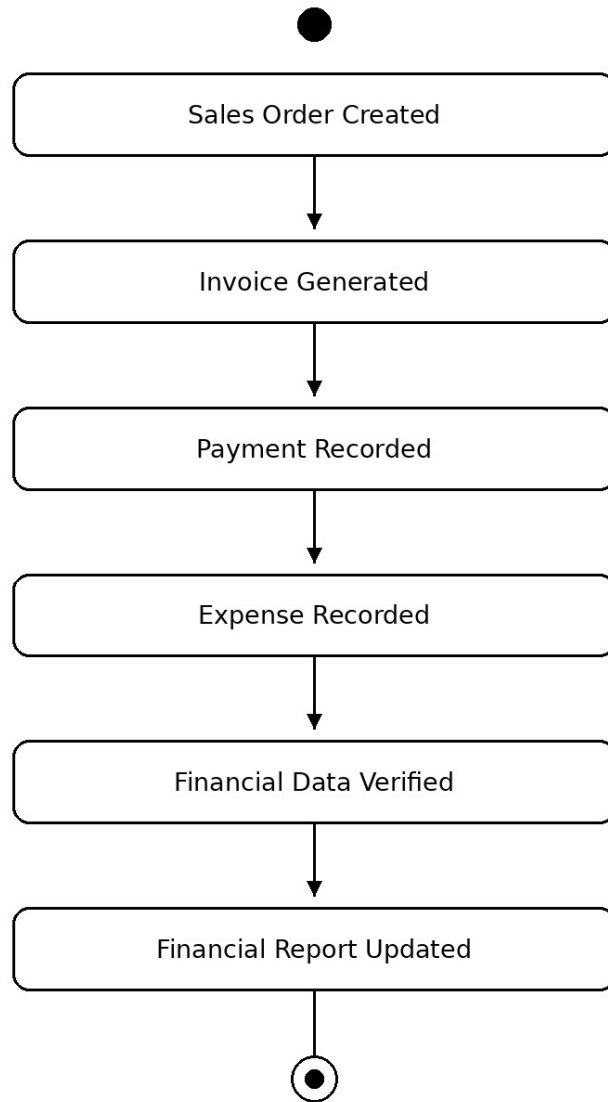


Figure 8: State Diagram for Sales, Billing and Expense Tracking

The financial workflow moves from sales order creation to invoice generation, payment recording, expense recording, verification, and financial-report update. A discrepancy can move the process to a review state.

9. Weather Monitoring and Alert:

- Weather Monitoring & Alert Management State Diagram

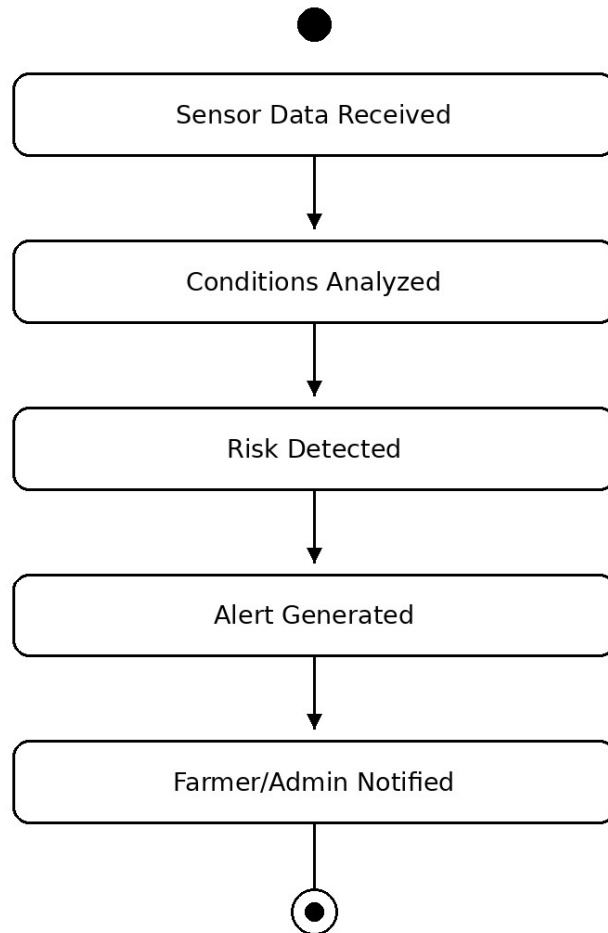


Figure 9: State Diagram for Weather Monitoring and Alert

The monitoring lifecycle moves from sensor data received to condition analysis, risk detection, alert generation, user notification, and monitored/cleared status.

3. Non-UML

Alongside the UML diagrams, several non-UML diagrams are used to represent the data structure, data movement, and overall architecture of the FarmFusion – Smart Agro Dairy Ecosystem. These diagrams provide a different perspective of the system and complement the UML design.

3.1 Entity-Relationship Diagram (ERD)

The Entity-Relationship Diagram represents the database structure of FarmFusion. It provides a clear view of the major entities, their attributes, and the relationships among them. The major entities include User, Farmer, Veterinarian, Staff, Livestock, HealthRecord, InventoryItem, FeedingSession, Crop, HarvestRecord, DairyProduct, WholesaleOrder, SalesOrder, Invoice, Payment, Expense, WeatherData, and Alert.

The User and role entities store account and access information. Livestock and HealthRecord store animal health information. InventoryItem and FeedingSession support resource and feeding management. Crop and HarvestRecord support cultivation and harvest activities. DairyProduct and WholesaleOrder support dairy commerce. SalesOrder, Invoice, Payment, and Expense support the financial workflow. WeatherData and Alert store environmental information and notification events.

The relationships between these entities show how operational data is connected within the system. For example, a farmer can manage livestock and crops, a livestock record can have multiple health records, a sales order can be linked with an invoice and payment, and weather data can trigger alerts. Overall, the ERD provides a structured foundation for data consistency, integrity, reduced redundancy, and organized database implementation.

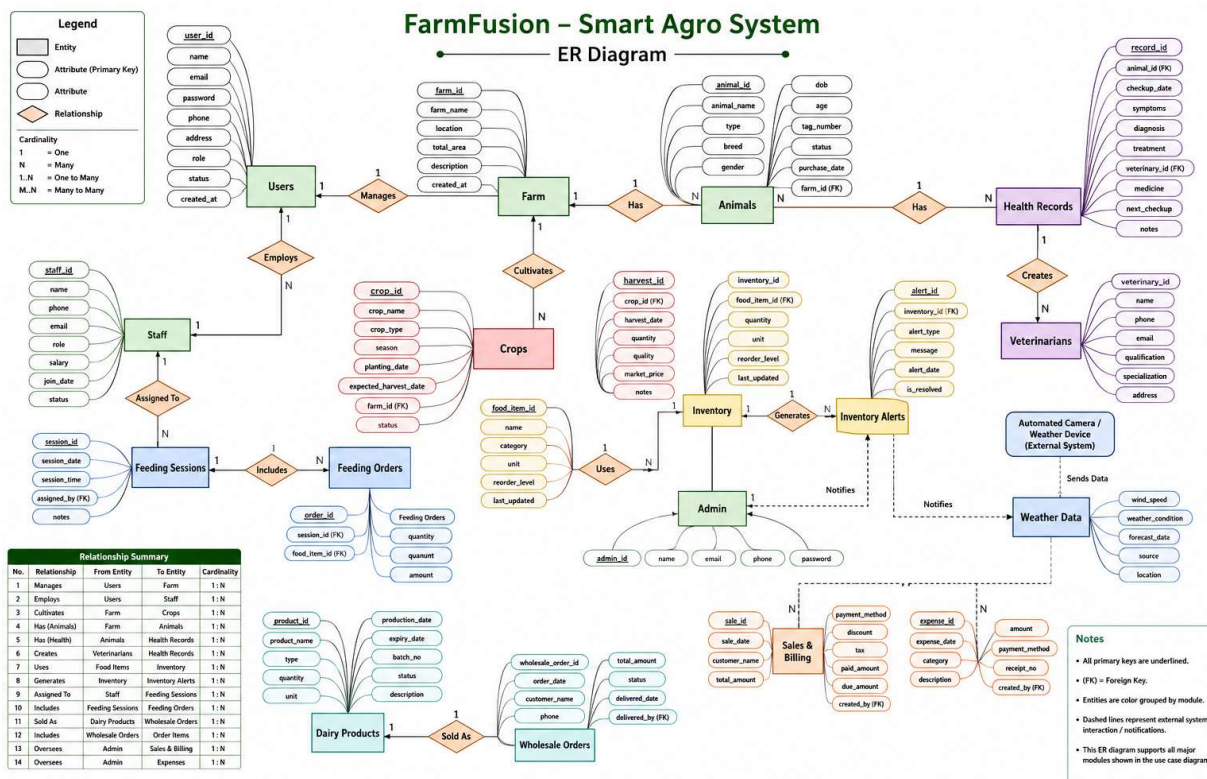


Figure 3.1.1: Entity Relationship Diagram of the FarmFusion – Smart Agro Dairy Ecosystem

The ERD (Figure 3.1.1) shows the main entities and relationships of the FarmFusion system and provides the database foundation for implementing the major farm-management modules.

3.2 Data Flow Diagram (DFD)

The Data Flow Diagram represents how information moves between external actors, FarmFusion processes, data stores, and automated services. The DFD complements the UML diagrams by focusing on data input, processing, storage, retrieval, and output.

Level 0

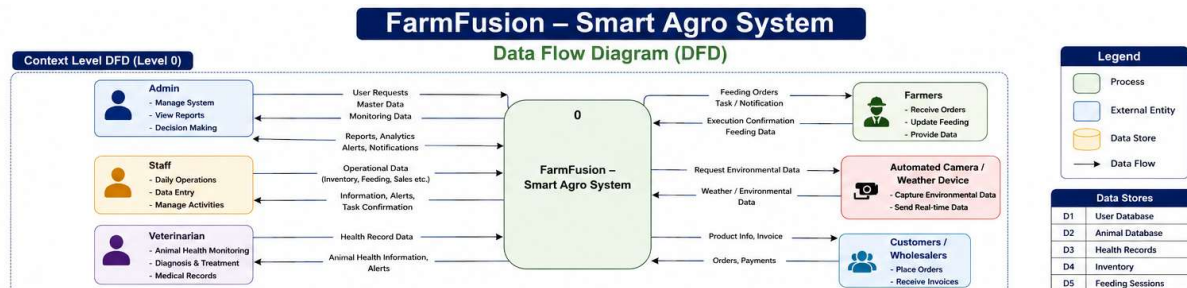


Figure 3.2.1: Level 0 DFD of the FarmFusion – Smart Agro Dairy Ecosystem

Level 0 shows the main data exchange between FarmFusion and its external actors, including Admin, Staff, Veterinarian, Farmer, Wholesaler/Customer, and environmental data sources. User information, operational records, orders, financial data, and weather information enter the system and produce dashboards, alerts, confirmations, and reports.

Level 1

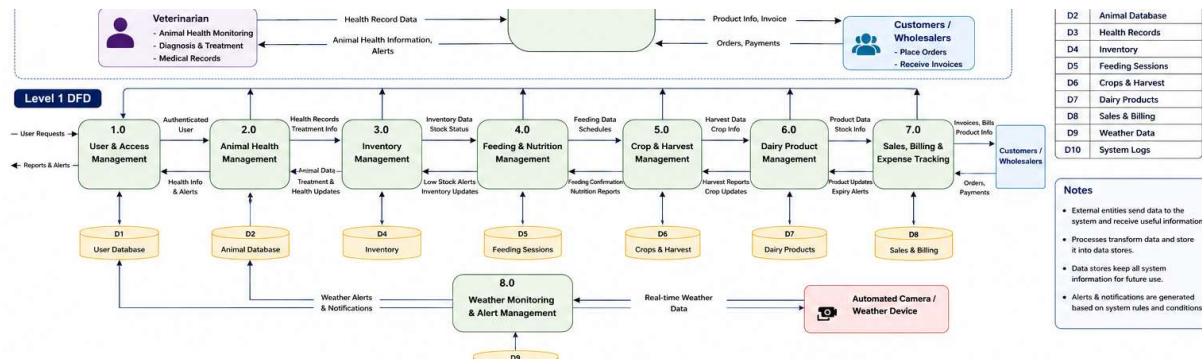


Figure 3.2.2: Level 1 DFD of the FarmFusion – Smart Agro Dairy Ecosystem

Level 1 decomposes the system into major processes such as authentication, animal health, inventory and feeding, dairy operations, wholesale, crop harvesting, sales and finance, and weather monitoring. It shows how these processes interact with their corresponding data stores and users.

Level 2

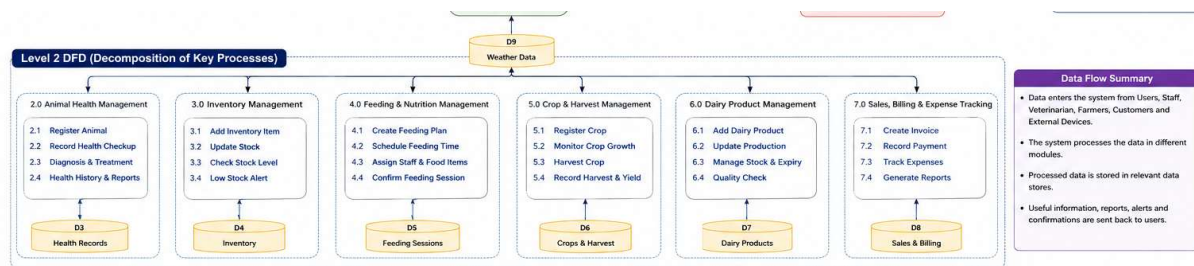


Figure 3.2.3: Level 2 DFD of the FarmFusion – Smart Agro Dairy Ecosystem

Level 2 gives a more detailed view of the internal processing of user-related and operational data. It demonstrates how information is validated, stored, retrieved, updated, and used to generate system responses and alerts.

3.3 Block Diagram

The Block Diagram shows the overall architecture of the FarmFusion system and the interaction between the user interface, application/business logic, database, automation and external data services. The architecture separates presentation, processing, persistent storage, and automated environmental inputs so that each layer can be maintained and expanded independently.

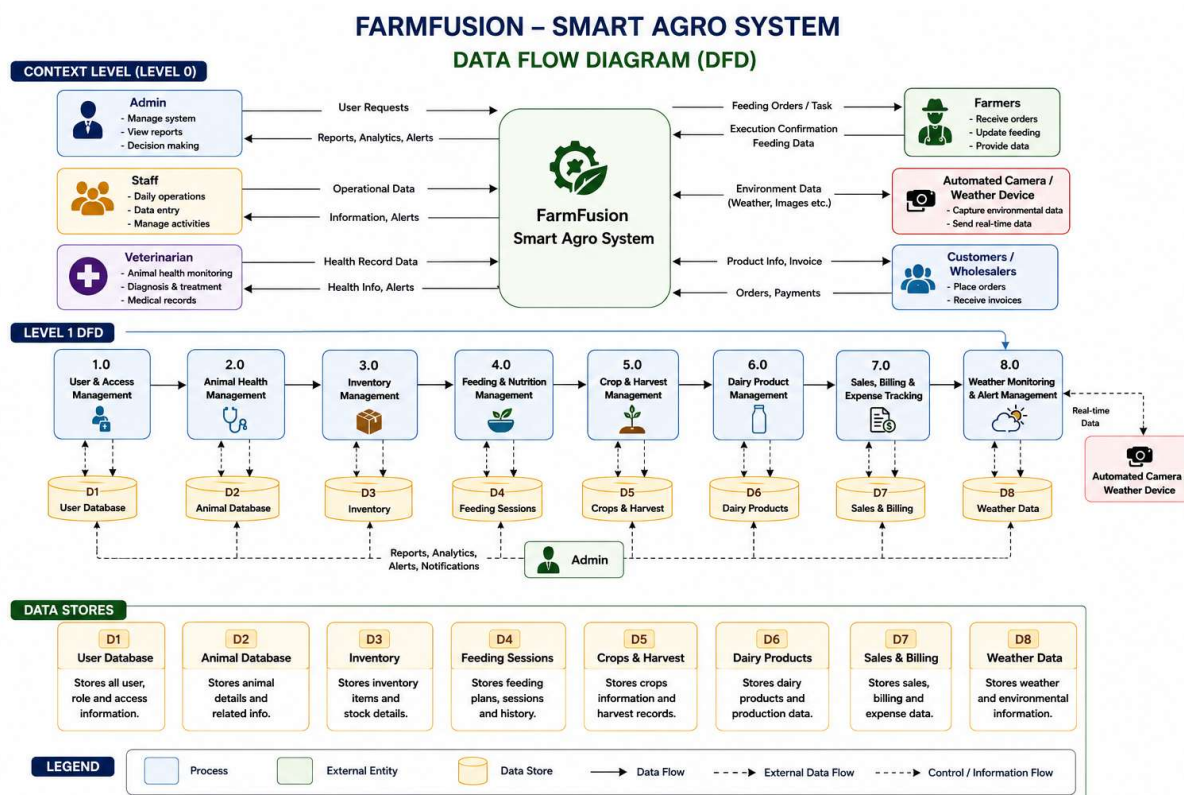


Figure 3.3.1: Block Diagram of the FarmFusion – Smart Agro Dairy Ecosystem

The Block Diagram (Figure 3.3.1) shows how users access the Frontend/UI, how the Backend/Application Layer processes requests and business rules, how the Database stores operational records, and how automated weather/environmental services provide data that can trigger alerts. This architecture provides a practical foundation for implementation.

4. Prototypes

An interactive and high-fidelity prototype was developed to evaluate the core architecture, functional pipelines, role-based controls, and user experience paradigms of the FarmFusion – Smart Agro Dairy Ecosystem. The prototype serves as a visual simulation layer that mirrors the role-oriented dashboards, operational modules, alerts, and live farm environment designed for the final implementation.

The implementation uses a responsive web interface with a consistent FarmFusion visual language. The prototype emphasizes role-based navigation, dashboard cards, operational command areas, animated farm scenes, status indicators, and module-specific workspaces. The screenshots below are taken from the developed FarmFusion prototype and demonstrate the actual interface rather than a generic mockup.

4.1 Client-Facing Landing Interface & System Navigation

The landing and authentication interface serves as the gateway for public interactions. It establishes the FarmFusion brand, introduces the smart-agro concept, and provides direct access to Sign In and Create Profile flows.

- **Core Design Matrix:** Uses a responsive split-screen layout with a visual farm environment and a focused authentication workspace.
- **Component Visual Elements:** Uses animated environmental elements, farm-operation highlights, role selection, secure sign-in controls, and a consistent dark-green visual system.
- **Navigation:** Provides clear entry points to registration and role-specific authentication.

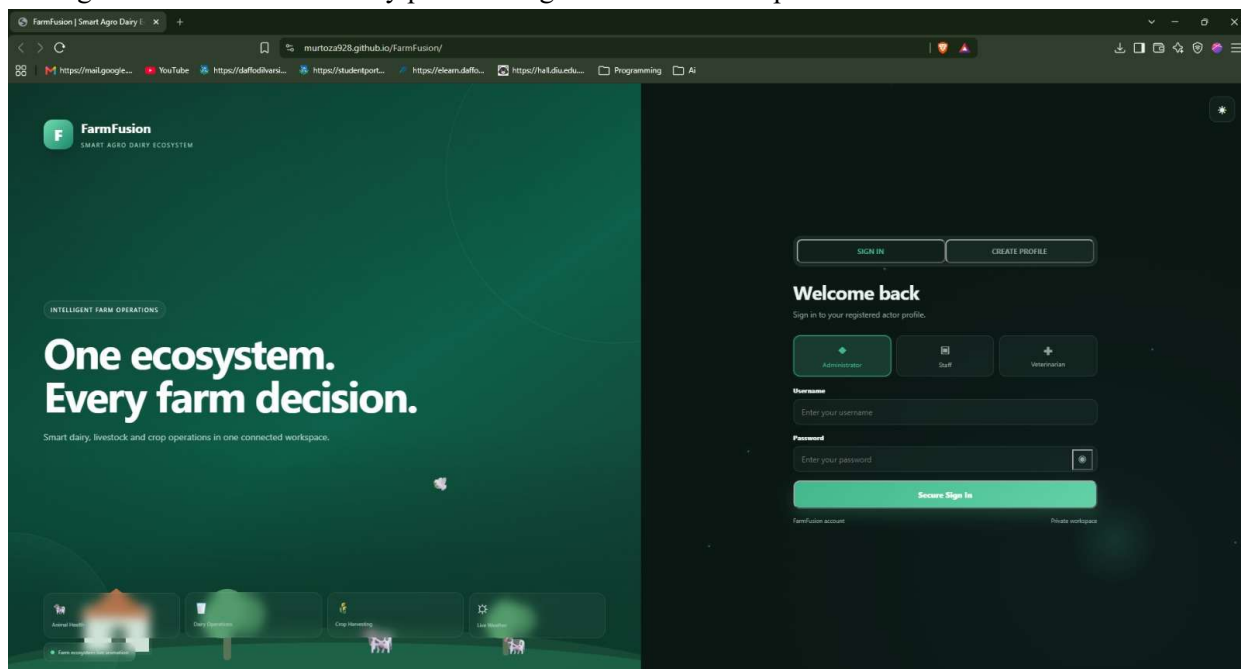


Figure 4.1.1: FarmFusion Landing and Authentication Interface

4.2 Role-Based Dashboard & Operational Command Console

After authentication, the user is redirected to a personalized dashboard according to the selected actor role. The dashboard provides an operational overview with animated farm visuals, key performance cards, module navigation, live activity, and quick-access actions.

- **Role-Based Navigation:** Different roles receive only the modules relevant to their responsibilities.

- **Dashboard Metrics:** The overview presents operational indicators such as animals, milk production, crops, revenue, expenses, profit, active staff, and inventory status.
- **Operational Console:** Users can move directly from the overview into Animal Health, Inventory, Dairy Products, Crop Harvesting, Sales/Billing/Expense Tracking, Weather Data, and Common Chat where permitted.

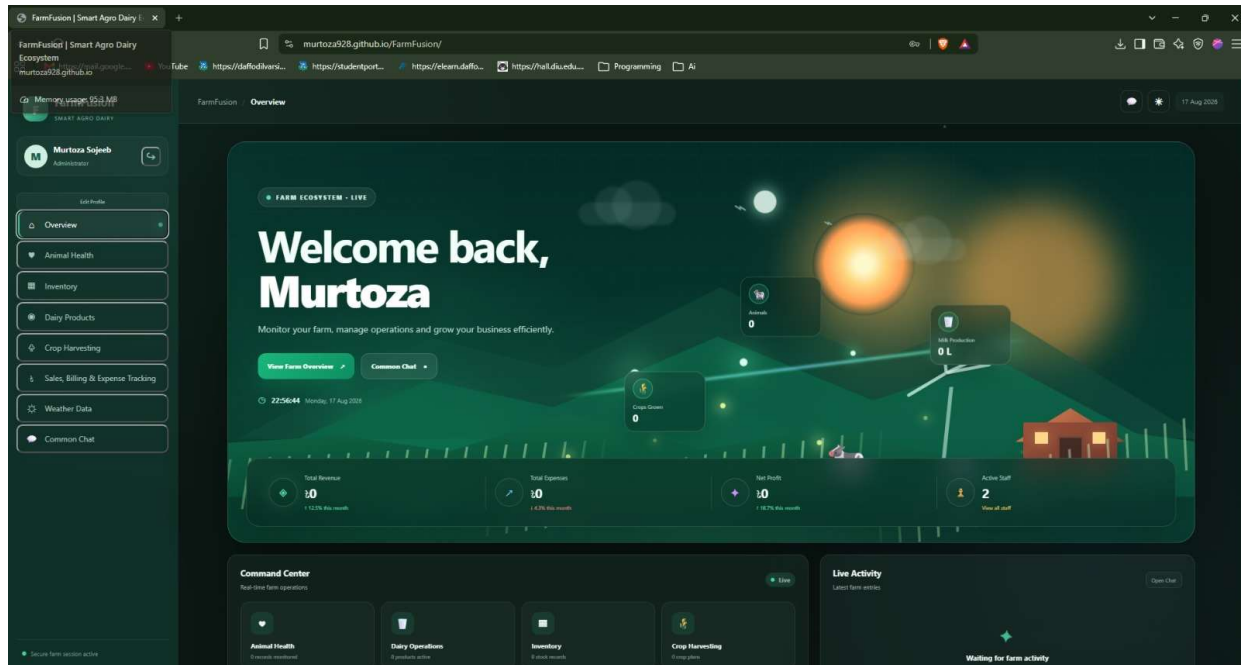


Figure 4.2.1: FarmFusion Administrator Operational Dashboard

4.3 Multi-Role Operational Workspace

The FarmFusion interface is designed to adapt to the responsibilities of each actor. Staff users receive a dairy-operations-oriented workspace, while the shared dashboard keeps common farm information visible without exposing unnecessary administrative controls.

- **Staff Workspace:** Focuses on dairy operations, inventory readiness, crop plans, products, and shared animal records.
- **Operational Pulse:** Uses progress indicators to communicate inventory readiness, dairy catalogue status, and farm activity.
- **Common Interaction:** The navigation and visual language remain consistent across roles to reduce learning time.

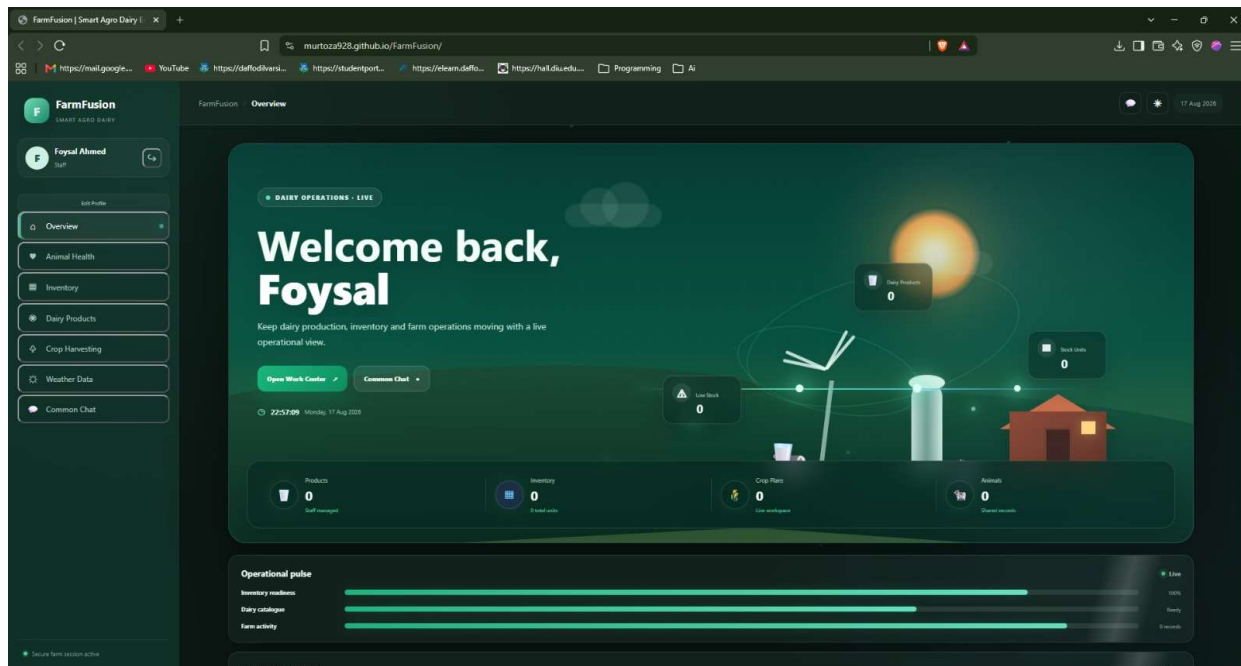


Figure 4.3.1: FarmFusion Staff Dashboard Prototype

4.4 Animal Health Monitoring & Veterinary Workspace

The veterinary workspace provides a focused environment for monitoring the farm's shared animal-health records. The interface emphasizes health signals, monitored animals, healthy counts, review queues, and average health indicators.

- **Clinical Pulse:** Shows the current health monitoring status and review queue.
- **Animal Health:** Provides direct access to animal-health records and updates.
- **Role Restriction:** The veterinarian dashboard is intentionally focused on Overview and Animal Health functions, matching the project's defined role boundary.

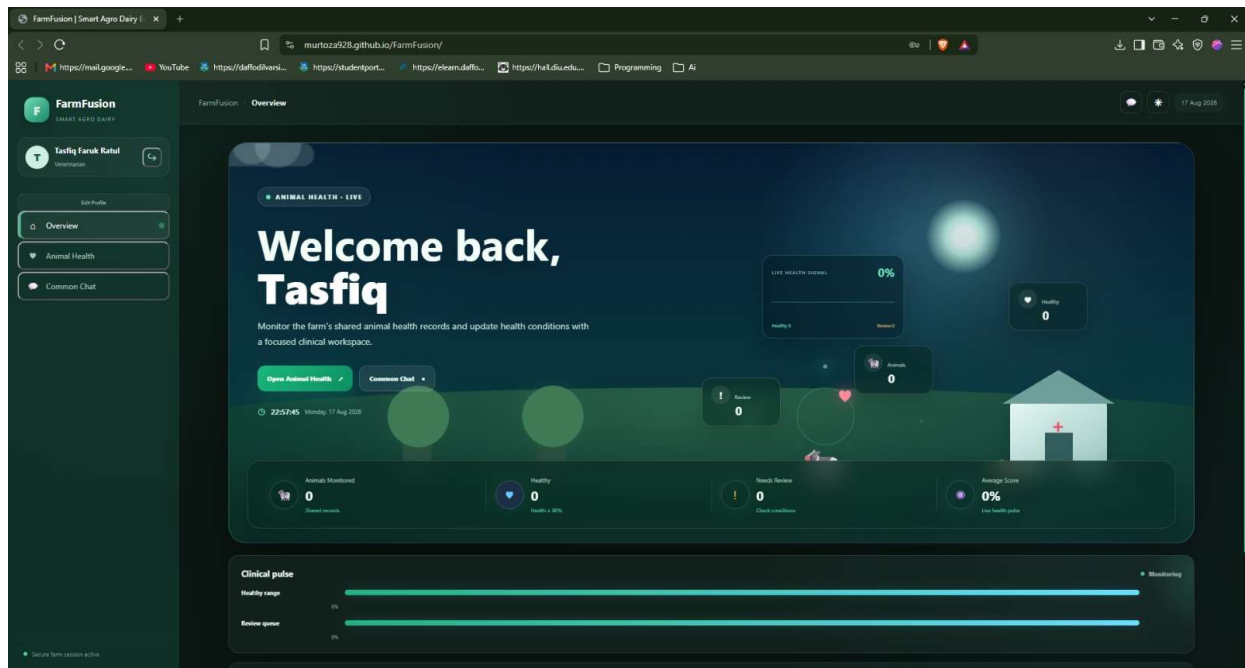


Figure 4.4.1: FarmFusion Veterinarian Animal Health Dashboard

4.5 Cross-Role Interface Consistency & Responsive Experience

The final prototype demonstrates a consistent visual and navigation system across the authentication, Administrator, Staff, and Veterinarian experiences. The same design language is maintained while the visible modules and dashboard content change according to the actor's responsibility.

- **Consistent Branding:** FarmFusion typography, iconography, dark-green visual language, animated farm scenes, and status indicators are shared across screens.
- **Responsive Layout:** The dashboard architecture is designed for large-screen web use and can be adapted to smaller displays.
- **User Experience:** Role-specific navigation reduces cognitive load while common interactions such as profile editing, theme control, notifications/chat, and dashboard navigation remain familiar.

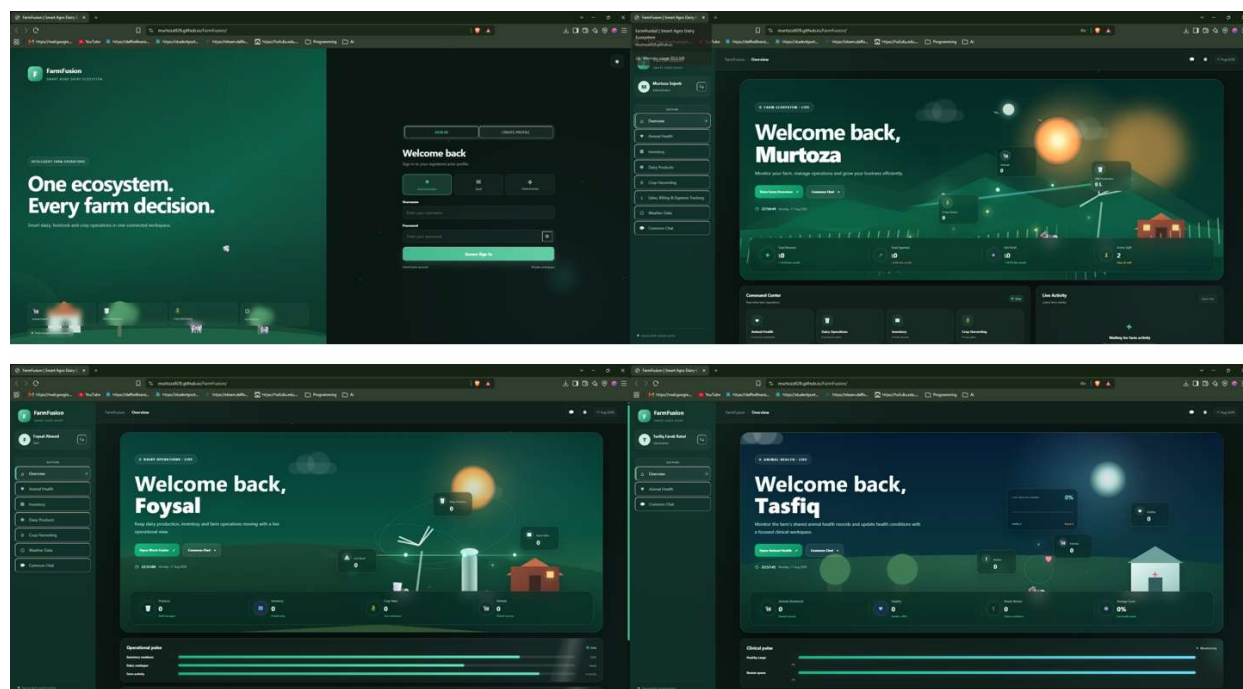


Figure 4.5.1: FarmFusion Cross-Role Prototype Comparison

5. Limitations

During the analysis and design phase of the FarmFusion – Smart Agro Dairy Ecosystem, several limitations were encountered. Identifying complete requirements, maintaining consistency among UML diagrams, defining relationships between operational entities, and representing complex farm processes were some of the major challenges.

Changes in system requirements can affect multiple diagrams simultaneously. For example, adding a financial workflow or changing role permissions may require corresponding changes in the Use Case Diagram, Class Diagram, Sequence Diagrams, Activity Diagrams, State Diagrams, ERD, DFD, and prototypes.

Another limitation was the difficulty of representing every possible user state and exceptional condition. Invalid inputs, unavailable inventory, failed transactions, incomplete health records, insufficient stock, unavailable weather data, and other alternative scenarios require additional consideration during detailed implementation.

The automated weather and environmental monitoring features also depend on the quality and availability of the connected data source. Similarly, automated camera/sensor functions require suitable hardware, network connectivity, calibration, and reliable data transmission.

Since some system components were designed before complete production deployment and field testing, certain technical, performance, usability, and operational aspects can only be evaluated more accurately after implementation with real farm users and real operational datasets.

Overall, the Design and Implementation documentation provides a structured foundation for developing FarmFusion while recognizing that some technical, operational, security, and user-related situations may require further refinement during future development.